

GlobalBridge Consulting Ltd.: Technical Design Report

CSIT606 Enterprise Network Design

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Section 1: Executive Summary

GlobalBridge Consulting Ltd. needs a secure, resilient and scalable IT infrastructure to support 750 employees across its Accra Headquarters and Kumasi Branch, and to accommodate the organisation's projected three-year growth. The proposed solution delivers reliable connectivity, secure separation of business functions, centralised business systems, enterprise security, monitoring, backup and disaster-recovery capabilities. The design targets 99.999% availability for core business services and keeps client financial data within Ghana, as required.

The proposed network uses a structured, layered architecture that lets the organisation expand without major redesign. Business departments are logically separated to improve security and control access to sensitive information. Users receive a 1 Gbps wired access baseline, supported by higher-capacity connections within the network infrastructure. Headquarters receives 1 Gbps primary Internet connectivity and the Branch 500 Mbps, with diverse secondary connections providing at least 50% of primary capacity so priority business services continue if a primary connection fails.

The infrastructure combines resilient network, server and storage technologies to limit disruption from individual equipment failures. Core business applications run on a two-node virtualisation platform, backed by dedicated ERP database servers and redundant shared storage. Critical information is protected through multiple backup methods, including local backup storage and encrypted tape copies held at a geographically separate location within Ghana. A warm-standby disaster-recovery approach adds further recovery capability, with the most critical services designed to a 30-minute recovery-time target and a 15-minute recovery-point target.

Security follows a defence-in-depth approach covering Internet connectivity, the internal network, wireless access, servers, endpoints and administrative access. Measures include next-generation firewall protection, encrypted site-to-site communications, network segmentation, controlled access between departments, strong user and administrator authentication, endpoint protection and centralised security monitoring. Governance is aligned principally with ISO/IEC 27001:2022 and the Ghana Data Protection Act, 2012 (Act 843), with PCI DSS applying wherever cardholder data enters scope.

The design directly addresses several critical business risks. Service-outage risk is reduced through redundant infrastructure, diverse WAN connectivity, backup and warm-standby disaster recovery. Cybersecurity and data-breach risk is reduced through layered security controls, endpoint protection, access control and centralised monitoring. Data-loss risk is addressed through the 3-2-1-1-0 backup strategy and geographically separated backup media. Compliance and data-residency risk is addressed by requiring client financial data to remain within Ghana and by making any service that cannot meet this requirement conditional on further validation. Procurement and physical-site uncertainty, finally, is managed through documented planning assumptions, required site-survey validation, procurement checks and retained budget headroom.

The final solution carries an estimated Grand Total of \$736,280.03, made up of \$431,696.87 CapEx and \$304,583.16 Year-1 OpEx. This is \$113,719.97, or 13.4%, below the \$850,000 budget ceiling. Because the initial cost solution exceeded the budget, documented value engineering was carried out.

Hardware-support coverage was aligned with equipment criticality, secondary WAN connections were correctly costed for their failover role, and the HQ access-switch platform was changed from Catalyst 9300 to the more cost-effective Catalyst 9200, since the additional capabilities of the higher platform were not required. A researched endpoint-security price was also used in place of the original planning estimate. These decisions brought the project within budget without reducing the required capacity, security, availability or recovery objectives.

The investment is expected to deliver business benefits through improved service continuity, reduced risk of costly downtime and data loss, stronger protection of client information, support for regulatory obligations and capacity for three-year organisational growth without major infrastructure redesign. Centralised monitoring and management should also improve operational visibility and incident response. These benefits provide the business justification for the investment, even where a direct monetary ROI cannot be calculated from the case-study information supplied.

The remaining \$113,719.97 budget headroom has deliberately not been spent simply to exhaust the available funding. Some final costs still depend on site surveys, supplier quotations and procurement validation, including physical telecommunications-room requirements, wireless placement, UPS sizing, WAN pricing and Ghana-based disaster-recovery services. Retaining this headroom protects the approved budget against reasonable implementation and procurement variation, while avoiding unnecessary expenditure on capabilities for which no current business requirement has been established.

Overall, the proposed solution gives GlobalBridge a secure, resilient, scalable and financially sustainable infrastructure that supports current operations and future growth while remaining within the approved budget and addressing the organisation's principal availability, security, data-protection and business-continuity risks.

Section 2: Requirements Analysis

2.1 Business Requirements

BR	Priority	Requirement	Business Driver	Infrastructure Implication
BR-001	Critical	The infrastructure must support the current workforce of 500 employees at HQ and 250 employees at the Branch from day one of operation.	GlobalBridge is establishing new dual-city operations and needs infrastructure sized to its actual current headcount at both sites, not a hypothetical or reduced population.	Switching, wireless, and server capacity at both sites must be provisioned for the full current workforce, addressed in Sections 3 and 4.
BR-002	High	The infrastructure must support the distinct operational and security needs of departments at both sites through appropriate separation and controlled communication.	GlobalBridge's departments (Finance, HR, Sales, Engineering, Marketing, Executive at HQ; Operations, Field Support, Regional Sales at Branch) handle information of differing sensitivity, and unrestricted lateral access between them would expose sensitive data unnecessarily.	Network segmentation and controlled inter-segment communication, addressed in Sections 3.3 and 5.2; administrative access is likewise restricted, addressed in Section 5.2.

BR	Priority	Requirement	Business Driver	Infrastructure Implication
BR-003	Critical	The infrastructure must provide 99.999% availability for core business systems.	Service interruption to core business systems directly affects GlobalBridge's ability to operate across both sites, so the business requires a high normal-operation availability target.	Redundant infrastructure design with no single point of failure, addressed in Sections 3 and 4; tiered recovery time/point objectives and the backup/DR mechanisms to achieve them when resilience alone is insufficient, addressed in Section 7 and TR-006; central monitoring / logging / administration supporting availability management, addressed in Section 6.
BR-004	Critical	The infrastructure must ensure that all client financial data remains within Ghana.	GlobalBridge holds client financial data subject to Ghana data-residency expectations, and operates in a regulatory environment governed by the Ghana Data Protection Act 2012, with ISO/IEC 27001 and (conditionally) PCI-DSS also relevant.	Storage, backup, replication, and recovery locations for client financial data must remain within Ghana, addressed in Sections 4, 5, and 7; the design must support the named compliance frameworks, addressed via TR-007/TR-008 and Section 5.5.
BR-005	Critical	Deliver the infrastructure within the approved USD \$850,000 budget envelope.	GlobalBridge has approved a fixed capital and first-year operating budget for this infrastructure programme and requires the design to be delivered within that envelope, or any excess explicitly justified.	Equipment, licensing, and services selection is tested against the \$850,000 ceiling, with trade-off decisions documented where needed, addressed in Section 8.
BR-006	Medium	The infrastructure must accommodate the specified three-year growth projections without requiring major redesign.	GlobalBridge's workforce and device population are expected to grow over a three-year horizon, and the business requires infrastructure that absorbs that growth incrementally rather than needing substantial rebuilding.	Addressing, switching, wireless, server, and storage capacity are each sized against three-year growth projections, addressed across Sections 3.4, 4.2, 4.3, and 8.

2.2 Technical Requirements

TR-001: Current-State Capacity. Switching and WAN capacity must support the current workforce at each site (500 HQ, 250 Branch) from day one.

TR-002: Three-Year Capacity. The addressing and switching architecture must accommodate the department-level three-year host growth projections derived in the project's VLSM addressing plan, so the network has sufficient capacity for three years of growth.

TR-003: Departmental Security Segmentation. The network must provide logical segmentation between departments, with communication between segments explicitly controlled rather than open by default.

TR-004: Availability (SLA). Core business systems must achieve a target availability of 99.999% (the target Service Level Agreement, SLA), equivalent to approximately 5 minutes 16 seconds of cumulative downtime per year.

TR-005: No Single Point of Failure. Paths and systems supporting core business services must not depend on a single device or link.

TR-006: Recovery Time and Recovery Point Requirements. Business services shall be classified into three recovery tiers according to operational criticality. Tier 1 critical services correspond to the core business systems subject to the availability requirement defined in TR-004. Tier 1 services shall have a Recovery Time Objective (RTO) of 30 minutes and a Recovery Point Objective (RPO) of 15 minutes. Tier 2 important services shall have an RTO of 4 hours and an RPO of 1 hour. Tier 3 standard services shall have an RTO of 24 hours and an RPO of 24 hours. TR-004 and TR-006 measure different aspects of service continuity: TR-004 defines the normal availability target, supported by the resilience requirement established in TR-005, while TR-006 defines the maximum recovery objectives for interruptions in which those normal resilience mechanisms prove insufficient. The detailed assignment of systems and services to these recovery tiers, together with the backup schedules, retention periods, disaster recovery procedures, and failover and restoration mechanisms required to achieve them, is defined in Section 7.

TR-007: Ghana Data Residency (Client Financial Data). All client financial data must be stored, backed up, replicated, and recovered only within Ghana.

TR-008: Regulatory and Standards Compliance. The infrastructure design must support compliance with Ghana Data Protection Act 2012, ISO/IEC 27001 controls, and PCI-DSS where payment-card data is in scope.

TR-009: Central Monitoring, Logging and Administration. Network infrastructure must support central monitoring, logging, and administration across both sites.

TR-010: Network Performance Requirements. The production LAN and WAN infrastructure shall provide sufficient bandwidth, throughput and latency to support GlobalBridge's 500 users at the Accra headquarters, 250 users at the Kumasi Branch, and projected three-year growth.

For the proposed production network, wired user access shall provide a minimum connection speed of 1 Gbps, while core, distribution and shared server uplinks shall provide at least 10 Gbps where traffic is aggregated. The production Internet service shall be sized initially at 1 Gbps symmetric for Accra HQ and 500 Mbps symmetric for the Kumasi Branch. Secondary connectivity should provide at least 50% of the primary link capacity to maintain priority business services if the primary connection fails.

Under normal operating conditions, critical network links should remain below 70% sustained utilisation to retain capacity for traffic bursts, failover and future growth. Internal LAN round-trip latency should normally remain at or below 5 ms, while HQ-to-Branch round-trip latency should target 50 ms or less.

For real-time voice traffic, one-way latency should remain at or below 150 ms, jitter should remain at or below 30 ms, and packet loss should not exceed 1%. These voice-performance thresholds are supported by authoritative industry guidance (ITU-T G.114 for voice latency; standard VoIP jitter/loss guidance).

These numerical values are design performance targets set for this project rather than figures supplied by the case study. They provide measurable criteria for network design, equipment selection, WAN procurement, monitoring and future capacity planning.

The Cisco Packet Tracer implementation demonstrates the required network architecture, configuration and connectivity rather than benchmarking production performance. Consequently,

simulated interface speeds may differ from the production performance targets without constituting a design failure, provided the difference is documented as a simulation limitation.

TR-011: Administrative Access Control. Administrative access to network infrastructure must be restricted to authorised management sources and use secure management protocols; unused access paths and interfaces must not remain available for unauthorised use.

2.3 Constraints

Group A: Business / Design Constraints

C-001 Budget. Grand Total (CapEx + Year-1 OpEx) $\leq$ USD \$850,000.

C-002 Ghana data residency. Client financial data must not be stored, backed up, replicated or recovered outside Ghana.

C-003 Missing physical building information. Physical building information, including floor plans, MDF/IDF locations and cable distances, is not supplied by the case study. Section 4.4's rack layout, cabling, and physical security design will state this limitation explicitly rather than inventing building specifics.

C-005 Prescribed address spaces. Production addressing must use the prescribed address spaces: 10.10.0.0/16 for HQ, 10.20.0.0/16 for Branch, and 172.16.0.0/12 for servers/DMZ.

C-009 Existing technology commitments. No existing technology commitments are specified in the case study. GlobalBridge's Accra and Kumasi operation is described as a new dual-city environment, so the infrastructure design proceeds on a greenfield basis, with no legacy hardware, incumbent platform or existing-system integration constraint identified in the supplied requirements.

Timeline. No infrastructure implementation timeline is specified in the supplied case study. Detailed implementation scheduling will require confirmation with GlobalBridge before deployment.

Group B: Assessment / Simulation Constraints

Production and simulation distinction (framing principle). Packet Tracer is used to demonstrate the required logical architecture, configuration and connectivity using the capabilities available in the simulation environment. Production recommendations are selected independently against GlobalBridge's business, technical, security, availability and capacity requirements. Where the production design differs from the Packet Tracer implementation, the difference shall be explicitly identified and justified rather than silently treated as equivalent.

A-001 Prescribed PT device families: retained as a Part B constraint only; does not determine Section 4 production equipment selection, which is justified independently per the framing principle above.

A-002 VTP implementation. The Packet Tracer implementation shall use the prescribed VTP configuration, including the domain name itinfra2026.local and the required server/client roles. In the project design, HQ and Branch operate as independent VTP instances because no Layer 2 trunk extends between the sites; inter-site connectivity is provided at Layer 3 through the simulated GRE path.

A-003 Native VLAN 99: retained; also flagged as a Part A/Part B consistency item.

A-004 LACP EtherChannel (Core-Distribution, src-dst-ip): retained as a Part B constraint; production link-aggregation use must be justified independently in Section 3/4.

A-005 Router-on-a-Stick for VLANs 40/50/90: retained as a Part B constraint. Production architecture may independently prefer full Layer 3 switching for inter-VLAN routing; this is a documented, intentional production-vs-simulation difference.

A-006 OSPF Area 0: retained as a Part B constraint. Section 3.5 provides a genuine OSPF vs. EIGRP vs. static comparison independent of this constraint.

A-008 GRE tunnel (site-to-site simulation): retained as a Part B constraint. GRE provides no encryption; GlobalBridge's production site-to-site security is independently selected and justified in Section 3.6.

A-009 TDR/Packet Tracer consistency. Elements intentionally represented by both deliverables, including VLAN IDs and names, IP addressing, gateways, routing relationships, ACL behaviour and other directly implemented logical design elements, shall remain consistent unless a production-versus-simulation difference is explicitly documented.

Section 3: Network Architecture and Design

3.1 Network Architecture Overview

GlobalBridge's production network will use a hierarchical enterprise architecture built on Core, Distribution and Access functions. This model suits the new dual-city environment because it separates network responsibilities into defined functional layers, allowing capacity, resilience, security and management controls to be introduced without treating the entire network as one large switching domain.

At the Access layer, wired and wireless users, phones and other endpoint devices connect to the network and are assigned to the appropriate departmental or service segment; this is where endpoint connectivity and access controls are applied. The Distribution layer aggregates access-layer connectivity and forms the policy boundary between user segments and the network core. The Core layer provides high-capacity connectivity between major network functions, server resources, WAN services and other parts of the enterprise infrastructure. This separation allows faults, configuration changes and future expansion to be managed more predictably than in a flat network.

The hierarchical approach suits the 500-user Accra headquarters because the larger site must support several departments, shared services and aggregated network traffic while allowing the infrastructure to grow without major redesign. The 250-user Kumasi Branch follows the same functional design principles but uses a smaller physical implementation appropriate to its lower user population, avoiding unnecessary equipment while preserving a consistent operational model between the two locations.

The architecture also supports GlobalBridge's three-year growth requirement. Access capacity can be increased by adding or expanding access switches without redesigning the addressing or routing architecture, while distribution and core capacity can be increased independently where traffic aggregation requires it. The production performance targets established in TR-010, including 1 Gbps wired endpoint connectivity and at least 10 Gbps for aggregated core, distribution and shared-server links, provide the capacity baseline against which production equipment will be selected.

Availability is another reason for choosing a hierarchical design. GlobalBridge requires a 99.999% SLA target for core business systems under TR-004, while TR-005 requires that core services not depend on a single device or link. The production architecture must therefore support redundant network paths and infrastructure, so that routine component or link failures can be handled without relying immediately on disaster recovery. The exact production switching, routing, power and equipment redundancy needed to implement this principle is selected and justified in the relevant later sections of the TDR.

The architecture also supports security and manageability. Departmental segmentation can be maintained at defined network boundaries, with communication between segments controlled according to security policy rather than allowed to pass laterally without restriction. Centralised monitoring, logging and administration can be applied consistently across both sites. This structure keeps troubleshooting, capacity monitoring and policy enforcement more manageable than a flat architecture as the organisation grows.

A flat network architecture was not selected because combining a large proportion of GlobalBridge's users and services into a less structured network would give weaker fault isolation, make policy enforcement harder, and become increasingly difficult to scale and manage as the organisation grows. A full spine-leaf architecture was also not chosen as the general campus model, because the supplied requirements describe a conventional enterprise environment rather than a large, east-west-traffic-intensive hyperscale data centre. Introducing spine-leaf across the entire environment would add design and operational complexity without a demonstrated requirement to justify it.

For the data-centre portion specifically, a dedicated spine-leaf fabric is not proposed at the current design scale. The production design instead uses a compact Core-attached architecture, consistent with the wider hierarchical network, in which the server and storage environment is treated as a distinct, high-priority network segment with appropriately sized and resilient connectivity to the enterprise Core. The project currently identifies nine required server roles, but the supplied requirements do not establish the high-volume east-west server traffic or hyperscale workload density that would justify the additional switching fabric and operational complexity of a spine-leaf architecture. This decision was revalidated once Section 4.2's server/storage sizing was complete (13 VMs across two hypervisor hosts, two physical ERP servers, one physical backup server), and it remains valid: the growth from 8 to 13 VMs was driven entirely by the Wazuh SIEM high-availability topology, not by new business workload, and does not change the underlying east-west traffic pattern.

The Packet Tracer simulation provides a simplified implementation of this concept. Its server VLAN 80 resources connect directly to HQ-CORE-SW1, but this simulated connection is evidence of the implemented Packet Tracer topology, not the sole justification for the proposed production data-centre architecture.

Cost is also a factor in the architecture. The hierarchical model lets GlobalBridge apply higher-capacity, resilient equipment where traffic and business criticality justify it, while using appropriately sized access infrastructure closer to users, an approach preferable to deploying the same high-cost platform throughout the network. Final equipment choices were tested against the \$850,000 project budget in Sections 4 and 8.

The design must also support GlobalBridge's Ghanaian operating requirements. In particular, network connectivity to server, storage, backup and disaster-recovery services must preserve the requirement that client financial data remain within Ghana. The architecture therefore provides the network foundation for the Ghana-based data placement and recovery controls specified in the server, security and disaster-recovery sections.

The Cisco Packet Tracer implementation is a functional simulation of this architecture rather than a complete representation of the proposed production infrastructure. Packet Tracer uses the devices and configuration mechanisms the simulation platform supports, so some implementation details differ from the production recommendation. For example, the simulation uses a Router-on-a-Stick arrangement for VLANs 40, 50 and 90 and currently operates some links at 100 Mbps, while the production architecture uses Layer 3 switching as the preferred inter-VLAN approach and is sized against the higher performance targets established in TR-010. These differences are documented simulation limitations, not production design choices.

3.2 Physical and Logical Topology Diagrams

Every diagram carries a common title block (project name, diagram name, site, version, date, drawn by), a consistent legend across all four figures (Core switching, Distribution switching, Access switching, WAN/edge router, Firewall, Servers/storage, User/end-device networks, Primary link, Redundant link, Trunk link, Routed link, WAN/service-provider boundary), and link annotations showing each link's speed, consistent with the production performance targets in TR-010 rather than the Packet Tracer simulated interface speeds where the two differ.

Figures 3.1 (HQ Campus Physical Topology) and 3.2 (Kumasi Branch Physical Topology) use labeled conceptual functional zones (e.g., “Server Room,” “Finance/HR User Area,” “Access Area,” “Communications Room”) rather than actual floor plans or building geometry, since no floor plans, MDF/IDF locations, or cable-run distances were supplied under C-003. Each carries the caveat: “Physical placement note: Actual building floor plans, telecommunications-room locations and cable-run distances were not supplied. Locations shown are conceptual functional zones used for design documentation.” Figure 3.3 (WAN/Inter-Site Connectivity) reflects the dual-transport Secure SD-WAN and site-to-site IPsec architecture established in Section 3.6. Figure 3.4 (Data Centre Logical Topology) reflects the Core-attached (non-spine-leaf) architecture established in Section 3.1 and revalidated against the final Section 4.2 server/storage estate.

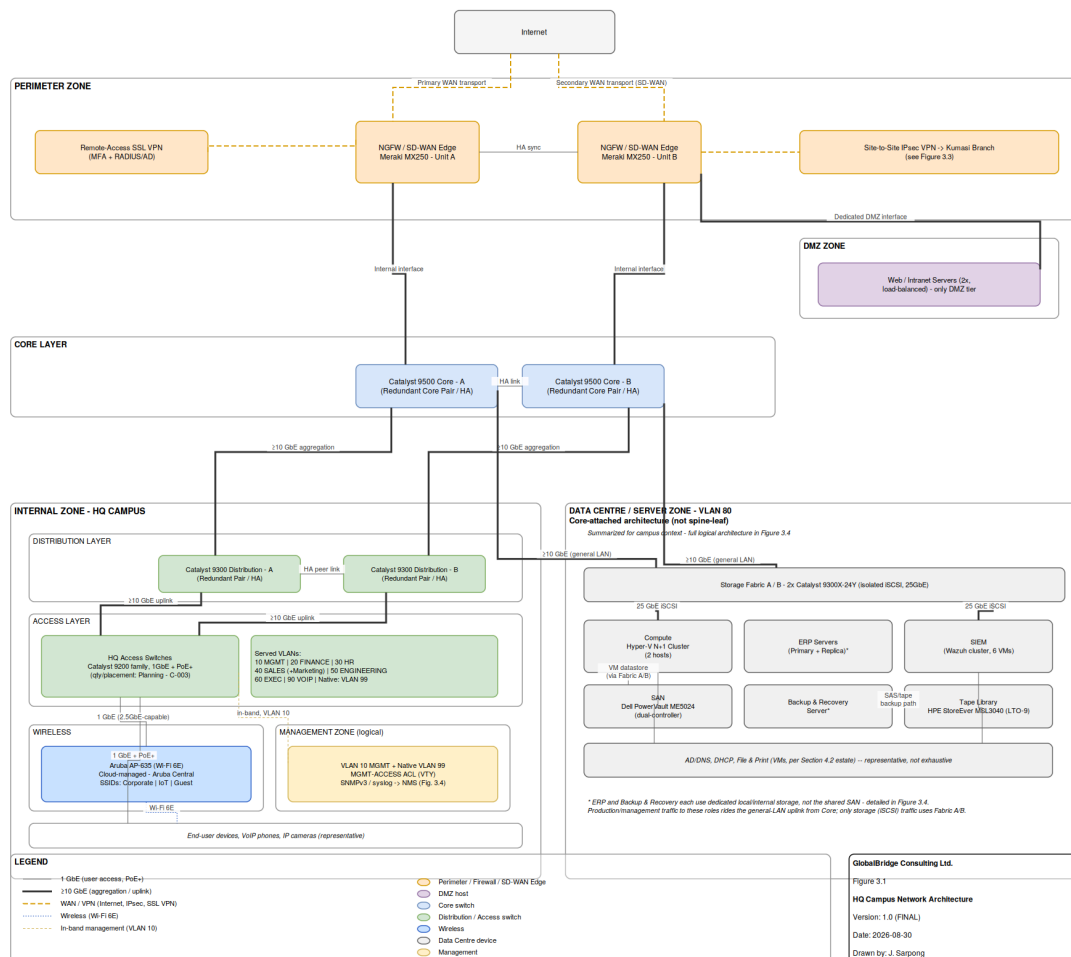


Figure 3.1: HQ Campus Network Architecture. Version 1.0 (FINAL), 2026-08-30. Drawn by J. Sarpong.

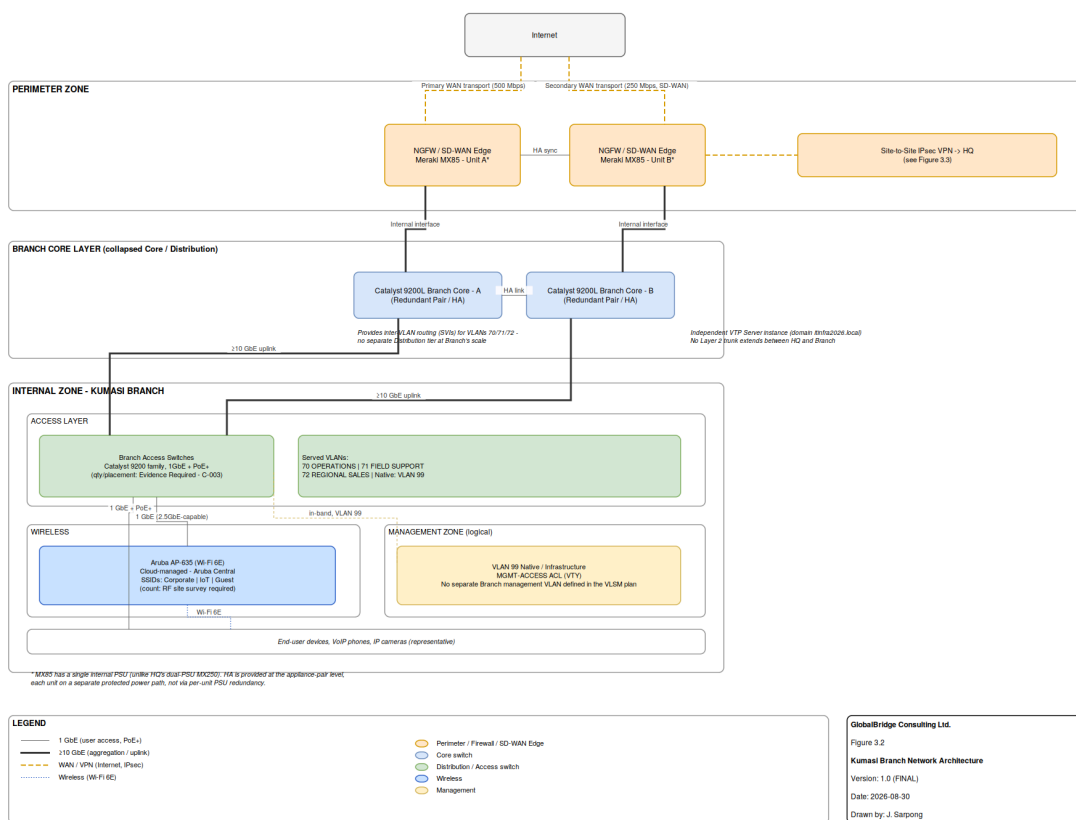


Figure 3.2: Kumasi Branch Network Architecture. Version 1.0 (FINAL), 2026-08-30. Drawn by J. Sarpong.

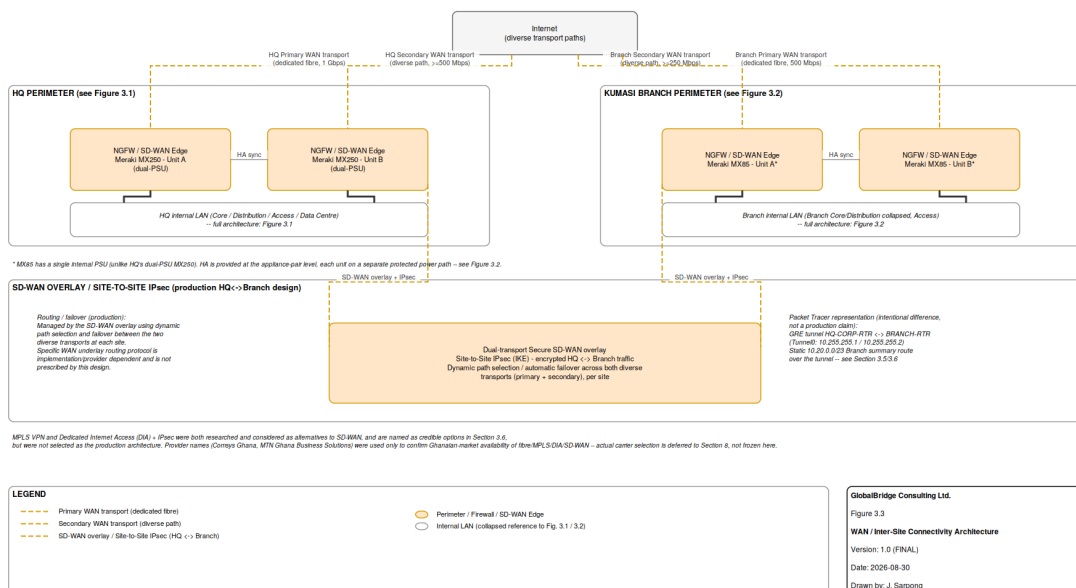


Figure 3.3: WAN / Inter-Site Connectivity Architecture. Version 1.0 (FINAL), 2026-08-30. Drawn by J. Sarpong.

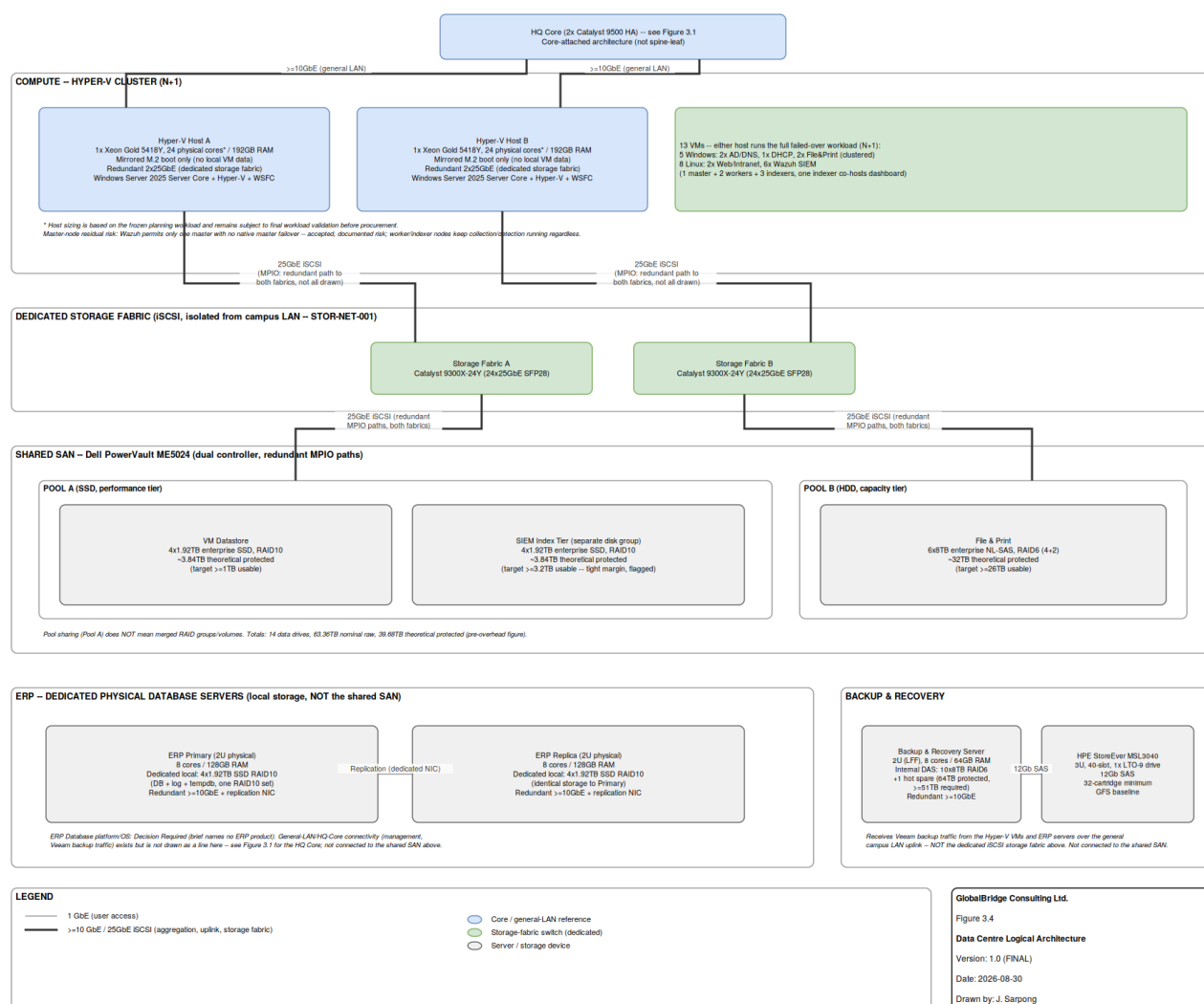


Figure 3.4: Data Centre Logical Architecture. Version 1.0 (FINAL), 2026-08-30. Drawn by J. Sarpong.

3.3 VLAN Design

GlobalBridge's VLAN scheme separates the network into departmental and functional broadcast domains at both sites, consistent with the hierarchical Core/Distribution/Access architecture adopted in §3.1 and the departmental segmentation requirement established in TR-003. Each VLAN corresponds to a distinct department, service, or administrative function, sized against the three-year host projections in the VLSM addressing plan (§3.4). Two VLANs, Servers (80) and Native/Management (99), use address ranges fixed by the case study rather than a calculated departmental allocation.

Marketing is named in the case study as an HQ department but is not given a dedicated headcount or VLAN requirement. Rather than invent an unsupported capacity figure, Marketing shares VLAN 40 (SALES) with the Sales department; this is a documented project assumption, not a figure given in the case study, and VLAN 40 keeps its SALES name throughout the TDR and the Packet Tracer build. VLAN 60 (EXEC) is scoped to Executive Management only; Operations is a Branch department (VLAN 70), not part of this VLAN. At the Branch, the case study gives a single combined headcount (250 current / 350 projected) across Operations, Field Support, and Regional Sales; this project's own assumption splits that total 60% / 16% / 24% respectively across VLANs 70/71/72, reconciling exactly to the combined figures. The resulting VLAN 71/72 device-level mapping (71 = FIELD_SUPPORT, 72 = REGIONAL_SALES) is confirmed by live show vlan brief evidence captured directly from BRANCH-CORE-SW1, not assumed.

Unless the table below states otherwise, every departmental VLAN's inter-VLAN boundary is controlled per TR-003 (explicit rather than open-by-default communication) and carries standard data traffic with no elevated QoS marking; the two exceptions, VoIP (90) and the TR-011 management path (10), are marked accordingly.

VLAN	Name	Site	Department / Purpose	3-Yr Hosts	Broadcast-Domain Justification	Boundary Security (exception to the TR-003 default above)	QoS (exception to the standard-data default above)
10	MGMT	HQ	Admin / management access to device management interfaces	40	Isolates admin traffic from user / department traffic	Per TR-011: authorised sources only, secure protocols, unused paths disabled	Standard control-plane handling
20	FINANCE	HQ	Finance department	160	Data-sensitive department, own broadcast domain	TR-003 default; Finance-to-HR restriction is a Section 5 decision	—
30	HR	HQ	Human Resources department	80	Separates HR's sensitive personnel data	TR-003 default; access policy in Section 5	—
40	SALES	HQ	Sales, and (by assumption) Marketing	300	Groups Sales with Marketing, which has no dedicated VLAN requirement	TR-003 default	—
50	ENGINEERING	HQ	Engineering department	200	Separates Engineering's traffic and systems	TR-003 default	—
60	EXEC	HQ	Executive Management	20	Smallest/most sensitive HQ segment	TR-003 default; same controlled-communication principle as Finance/HR	—

VLAN	Name	Site	Department / Purpose	3-Yr Hosts	Broadcast-Domain Justification	Boundary Security (exception to the TR-003 default above)	QoS (exception to the standard-data default above)
80	SERVERS	HQ	Shared infrastructure/application servers (172.16.0.0/24, prescribed)	20–50	Dedicated, tightly controlled segment, separate from end-user VLANs	Restricted from open user-VLAN access; zone/DMZ model in Section 5	Subject to TR-010 aggregation-capacity requirements; no VLAN-specific threshold
90	VOIP	HQ	Voice/telephony traffic	300	Separates real-time voice from data traffic	TR-003 default	TR-010 thresholds: latency ≤150 ms, jitter ≤30 ms, loss ≤1%
99	NATIVE/ MGMT	HQ	Native VLAN, trunk links; switch management SVIs (192.168.1.0/24, per exam spec)	—	Separates native / infrastructure-management traffic from user VLANs	Infrastructure-level only, distinct from VLAN 10; no other production use	Standard control-plane handling
70	OPERATIONS	Branch	Operations department	150 / 210	Branch's largest department (60% of combined headcount, by assumption)	TR-003 default	—
71	FIELD_SUPPORT	Branch	Field Support department	40 / 55	16% of combined Branch headcount, by assumption	TR-003 default; 71/72 mapping confirmed by Packet Tracer evidence	—

VLAN	Name	Site	Department / Purpose	3-Yr Hosts	Broadcast-Domain Justification	Boundary Security (exception to the TR-003 default above)	QoS (exception to the standard-data default above)
72	REGIONAL_SALES	Branch	Regional Sales department	60 / 85	24% of combined Branch headcount, by assumption	TR-003 default; 71/72 mapping confirmed by Packet Tracer evidence	—
99	NATIVE	Branch	Native VLAN, Branch trunk links	—	Same rationale as HQ VLAN 99	Same as HQ VLAN 99	Standard control-plane handling

(“3-Yr Hosts” for Branch rows shows current / three-year-projected; “—” in Boundary Security or QoS means the TR-003/standard-data default above applies with no exception.)

Every VLAN boundary in this table establishes a Layer 3 segmentation point at which inter-VLAN traffic is subject to explicit control rather than open by default, per TR-003. The specific mechanisms that enforce this at each boundary, access-control lists, firewall zone policy, and related controls, are covered in Section 5’s security architecture and are not exhaustively defined here; the Packet Tracer implementation’s BLOCK-FINANCE-TO-HR and MGMT-ACCESS ACLs demonstrate this principle for two specific boundaries in the simulation and are not presented as the complete production security policy for every VLAN listed above.

3.4 IP Addressing Plan (VLSM)

This section presents the complete IP addressing plan for GlobalBridge’s production network, derived using Variable Length Subnet Masking (VLSM) from the three address ranges given for this project: 10.10.0.0/16 for HQ, 10.20.0.0/16 for the Kumasi Branch, and the prescribed 172.16.0.0/12 address space for servers/DMZ. For each VLAN, the required subnet size is calculated from the three-year host projection established in §3.3, using the smallest prefix length satisfying $2^n - 2 \geq$ required hosts. Subnets are then allocated largest-block-first within their assigned supernet, so each block starts on a boundary aligned to its own size: the standard VLSM discipline used to avoid address fragmentation. Because Gateway and First-Usable Address are identical values by design in every subnet in this plan (the gateway is always the first usable host address), the two are presented as a single merged column rather than as separate, duplicated columns.

3.4.1 HQ Addressing (10.10.0.0/16)

Table 3.4.1a: HQ subnet allocation and capacity

VLAN	Name	Network Address	Subnet Mask (CIDR)	Subnet Mask (Dotted Decimal)	Host Capacity	3-Year Projected Hosts
40	SALES	10.10.0.0	/23	255.255.254.0	510	300

VLAN	Name	Network Address	Subnet Mask (CIDR)	Subnet Mask (Dotted Decimal)	Host Capacity	3-Year Projected Hosts
90	VOIP	10.10.2.0	/23	255.255.254.0	510	300
20	FINANCE	10.10.4.0	/24	255.255.255.0	254	160
50	ENGINEERING	10.10.5.0	/24	255.255.255.0	254	200
30	HR	10.10.6.0	/25	255.255.255.128	126	80
10	MGMT	10.10.6.128	/26	255.255.255.192	62	40
60	EXEC	10.10.6.192	/27	255.255.255.224	30	20

Table 3.4.1b: HQ gateway, usable range and broadcast

VLAN	Name	Gateway / First Usable	Last Usable	Broadcast Address
40	SALES	10.10.0.1	10.10.1.254	10.10.1.255
90	VOIP	10.10.2.1	10.10.3.254	10.10.3.255
20	FINANCE	10.10.4.1	10.10.4.254	10.10.4.255
50	ENGINEERING	10.10.5.1	10.10.5.254	10.10.5.255
30	HR	10.10.6.1	10.10.6.126	10.10.6.127
10	MGMT	10.10.6.129	10.10.6.190	10.10.6.191
60	EXEC	10.10.6.193	10.10.6.222	10.10.6.223

3.4.2 Servers and Infrastructure

The Servers VLAN (80) uses the prescribed server address space, while VLAN 99 uses the project's fixed infrastructure/native-management allocation. Neither is sized as a departmental user subnet.

VLAN	Name	Network Address	Subnet Mask (CIDR)	Subnet Mask (Dotted Decimal)	Gateway	Usable Range	Broadcast	Hosts
80	SERVICES	172.16.0.0	/24	255.255.255.0	172.16.0.1	172.16.0.1 – 172.16.0.254	172.16.0.255	20–50

VLAN	Name	Network Address	Subnet Mask (CIDR)	Subnet Mask (Dotted Decimal)	Gateway	Usable Range	Broadcast	Hosts
99	NATIVE/MGMT	192.168.1.0	/24	255.255.255.0	192.168.1.1	192.168.1.1 – 192.168.1.254	192.168.1.255	— (native/trunk-management use only)

3.4.3 Branch Addressing (10.20.0.0/16)

The case study gives the Kumasi Branch a single combined headcount, 250 current, 350 three-year projected, across its three departments (Operations, Field Support, Regional Sales) rather than a per-department figure. This project's own documented assumption splits that combined total 60% / 16% / 24% respectively across VLANs 70 (Operations), 71 (Field Support), and 72 (Regional Sales), reconciling exactly to the combined figures: 210 + 55 + 85 = 350 projected, 150 + 40 + 60 = 250 current. This split is an assumption made to provide three distinct Branch VLANs (Part B), not a figure supplied by the case study, and is stated here explicitly on that basis.

Table 3.4.3a: Branch subnet allocation and capacity

VLAN	Name	Network Address	Subnet Mask (CIDR)	Subnet Mask (Dotted Decimal)	Host Capacity	Current Hosts	3-Year Projected Hosts
70	OPERATIONS	10.20.0.0	/24	255.255.255.0	254	150	210
72	REGIONAL_SALES	10.20.1.0	/25	255.255.255.128	126	60	85
71	FIELD_SUPPORT	10.20.1.128	/26	255.255.255.192	62	40	55
	Total					250	350

Table 3.4.3b: Branch gateway, usable range and broadcast

VLAN	Name	Gateway / First Usable	Last Usable	Broadcast Address
70	OPERATIONS	10.20.0.1	10.20.0.254	10.20.0.255
72	REGIONAL_SALES	10.20.1.1	10.20.1.126	10.20.1.127
71	FIELD_SUPPORT	10.20.1.129	10.20.1.190	10.20.1.191

3.4.4 Representative VLSM Calculations

The following three worked examples illustrate the method applied consistently across every subnet in this plan.

SALES (VLAN 40): 300 three-year hosts. $2^8 - 2 = 254$, insufficient for 300 hosts. $2^9 - 2 = 510$, sufficient. Minimum prefix: /23 (255.255.254.0), allocated at 10.10.0.0/23.

HR (VLAN 30): 80 three-year hosts. $2^6 - 2 = 62$, insufficient for 80 hosts. $2^7 - 2 = 126$, sufficient. Minimum prefix: /25 (255.255.255.128), allocated at 10.10.6.0/25.

EXEC (VLAN 60): 20 three-year hosts. $2^4 - 2 = 14$, insufficient for 20 hosts. $2^5 - 2 = 30$, sufficient. Minimum prefix: /27 (255.255.255.224), allocated at 10.10.6.192/27.

3.4.5 Address-Space Scalability

After the allocations above, the range 10.10.7.0 – 10.10.255.255 remains unallocated within the HQ 10.10.0.0/16 supernet. This space is available for future VLAN growth beyond the current three-year projections, for example, a future 10.10.7.0/24 block, without requiring any change to the addressing already allocated above. No specific future block is formally reserved at this stage; the space is stated here as available capacity, not as a committed allocation.

3.4.6 TDR and Packet Tracer Addressing Consistency

The production subnet and gateway values represented in the Packet Tracer implementation are consistent with the addressing plan above. Depending on the VLAN, these values appear on SVIs, Router-on-a-Stick subinterfaces, static/routed interfaces, or DHCP scopes as applicable. The simulation uses only a small number of representative hosts per user VLAN while retaining the full-sized production subnets: a subnet showing a small number of active hosts against a much larger capacity is expected simulation behaviour, not an addressing discrepancy between the TDR and the build.

3.5 Routing Design

This section specifies and justifies GlobalBridge's production routing strategy, covering interior gateway protocol selection, inter-VLAN routing at both sites, inter-site routing, route summarisation, and default-route propagation.

3.5.1 Routing Protocol Selection

OSPF is selected as GlobalBridge's Interior Gateway Protocol (IGP), in preference to EIGRP and static routing, evaluated against three criteria: environment size, vendor mix, and operational complexity. On size, the HQ and Branch topologies described in this design are within the scale OSPF is designed for, requiring only a single Area 0 at the current design scope. On vendor mix, OSPF provides broader multi-vendor interoperability and avoids unnecessary dependence on a vendor-associated routing protocol; since the final equipment selection had not yet been established when this routing decision was made, the open-standard and widely interoperable option was preferred. On operational complexity, OSPF and EIGRP are comparable at this design's current single-area scale, so operational complexity did not independently favour either protocol; OSPF was selected on the size and vendor-mix reasoning above. Static routing alone was not selected as the production IGP, since it does not scale operational administration to a growing, multi-site network the way a dynamic IGP does. Static routing is, however, used deliberately at specific points in this design (§3.5.4) where a dynamic routing relationship is not required.

3.5.2 HQ Inter-VLAN Routing

HQ inter-VLAN routing uses a hybrid model, consistent with the Part B configuration requirements. Switched Virtual Interfaces (SVIs) on the Core Layer 3 switch provide inter-VLAN routing for VLANs 10 (MGMT), 20 (FINANCE), 30 (HR), 60 (EXEC), 80 (SERVERS), and 99 (NATIVE/MGMT). Router-on-a-Stick subinterfaces on the HQ WAN/Edge router provide inter-VLAN routing for VLANs 40 (SALES), 50 (ENGINEERING), and 90 (VOIP). This hybrid split follows the Part B configuration tasks. The resulting routes for the SVI-routed VLANs are learned by the WAN/Edge router through OSPF advertisements from the Core Layer 3 switch, demonstrating that the two inter-VLAN routing mechanisms interoperate rather than operating as isolated routing domains.

3.5.3 Branch Inter-VLAN Routing

Branch inter-VLAN routing uses SVIs on the Branch Core switch for all three Branch VLANs (70 OPERATIONS, 71 FIELD_SUPPORT, 72 REGIONAL_SALES): no Router-on-a-Stick arrangement is required at the Branch, since the Part B tasks do not call for one there. This keeps the Branch's inter-VLAN routing simpler than HQ's, appropriate to the Branch's smaller scale, while remaining architecturally consistent with the hierarchical model adopted in §3.1.

3.5.4 Inter-Site Routing

Inter-site reachability between HQ and the Branch is provided by a manually configured static summary route, not by OSPF. This is confirmed by the actual routed configuration: the Branch network appears in the HQ WAN/Edge router's routing table as a static route, and the GRE tunnel transport network is not included among the networks configured under the OSPF process. There is no OSPF adjacency formed across the GRE tunnel. The GRE tunnel itself is a Packet Tracer simulation mechanism only, standing in for the production WAN transport specified independently in §3.6; it carries no encryption and is not presented as GlobalBridge's production inter-site security mechanism.

3.5.5 Route Summarisation

Route summarisation is applied where the currently allocated address space permits a clean, size-aligned summary boundary, and is stated plainly where it is not yet configured. At the Branch, the three currently allocated department subnets (10.20.0.0/24, 10.20.1.0/25, 10.20.1.128/26) are exactly covered by the summary 10.20.0.0/23, and this summary is the actual static route configured for inter-site reachability (§3.5.4): summarisation here is both addressable and implemented. At HQ, the currently allocated departmental address space through 10.10.6.223 can be represented within the aggregate 10.10.0.0/21. However, that aggregate route is not configured in the current Packet Tracer implementation; the relevant HQ networks are carried individually. This distinction between addressing that could support future summarisation and summarisation that is actually configured is stated explicitly rather than overclaiming a summarisation posture the current design does not implement.

3.5.6 Default Routing

The HQ WAN/Edge router carries a static default route toward the simulated ISP, redistributed into OSPF so that HQ's internally routed devices (including the Core switch and its downstream SVIs) learn a default path to the Internet/WAN edge without each device requiring its own independently configured default route. The simulated ISP router carries its own static default route, consistent with its role as a simulation boundary rather than a production Internet edge. This default-route behaviour satisfies the Part B simulation requirement. In production, Internet and inter-site path selection would be implemented through the resilient WAN/SD-WAN architecture defined in §3.6 rather than assumed to reproduce the Packet Tracer edge-routing arrangement identically.

3.5.7 Routing, Security and Resilience

OSPF supports the production resilience strategy by providing dynamic route calculation and the ability to reconverge when alternative routed paths are available. It therefore complements the redundant-device and redundant-link architecture required by TR-005, but does not by itself remove a single point of failure. The Packet Tracer implementation demonstrates the required routing behaviour at simulation scale; the fuller production resilience design depends on the redundant infrastructure and WAN paths described elsewhere in the TDR. Routing itself does not enforce departmental security policy: the VLAN boundaries established in §3.3 are the segmentation mechanism, and the specific controls governing which VLANs may communicate across those boundaries (access-control lists, firewall policy) are defined in Section 5's security architecture, not created by the routing design on its own.

3.6 WAN & Internet Connectivity

This section specifies GlobalBridge's WAN and Internet connectivity strategy for both the Accra HQ and Kumasi Branch, covering primary and secondary transport, inter-site connectivity, and Internet breakout.

3.6.1 WAN Technology Evaluation

Three WAN technologies were evaluated for GlobalBridge's HQ-to-Branch and Internet connectivity: traditional MPLS VPN, dedicated Internet access (DIA) paired with site-to-site IPsec, and SD-WAN overlaying diverse Internet transports. Fortinet's published comparison of SD-WAN and MPLS was used as one technical reference for the characteristics and trade-offs considered in this evaluation. MPLS offers a carrier-managed, predictable-performance private WAN, but its per-site circuit cost and comparatively slower provisioning are weighed against a two-site environment that does not yet demonstrate a scale requiring MPLS's centralised traffic-engineering strength. DIA with IPsec provides direct Internet access at each site with encrypted inter-site tunnels, but without SD-WAN's centrally managed dynamic path selection across multiple transports. Comsys Ghana's and MTN Ghana Business Solutions' published service offerings confirm that fibre, MPLS, DIA, and SD-WAN transport options are each genuinely available in the Ghanaian market: cited here as market-availability evidence, not as technical proof that one architecture is superior.

3.6.2 Recommended Production WAN

The recommended production WAN architecture is dual-transport Secure SD-WAN over diverse Internet paths, a business-grade dedicated fibre connection as the primary transport at each site, with a diverse secondary path, combined with site-to-site IPsec (IKE) protecting all HQ-to-Branch traffic carried over the overlay. RFC 6071's IPsec/IKE standards roadmap is cited as the basis for this security mechanism. MPLS and DIA-with-IPsec were both evaluated and are not selected as the sole production architecture, in favour of SD-WAN's centrally managed path selection and failover flexibility, combined with site-to-site IPsec/IKE for protection of inter-site traffic: SD-WAN provides path management and resilience, while IPsec/IKE provides inter-site confidentiality and integrity; the two are distinct mechanisms, not one technology supplying both roles. MPLS and DIA-with-IPsec both remain named here as credible alternatives, not as rejected-and-discarded options. Actual carrier selection among the providers confirmed available in §3.6.1 is explicitly deferred to Section 8. This section fixes the WAN architecture and security mechanism, not a specific vendor contract.

3.6.3 Provider Diversity and Resilience

WAN resilience is achieved through diverse transport paths rather than reliance on a single provider's stated availability figure. A provider's own advertised SLA, for example, a figure in the high-99% range for a single circuit, is distinct from GlobalBridge's own 99.999% availability target for core business systems (TR-004): no single provider's SLA, however strong, satisfies TR-004 on its own, since TR-005 requires that core services not depend on a single device or link. The

dual-transport design in §3.6.2 is the architectural response to this: primary and secondary paths are diverse, so that a failure or degradation on one transport does not remove WAN connectivity outright. Where commercially and technically feasible, the primary and secondary transports should use different service providers and physically diverse last-mile paths, so that a single carrier or access-circuit failure does not defeat both WAN connections.

3.6.4 Inter-Site Security

HQ-to-Branch traffic is protected by site-to-site IPsec (IKE), carried over the SD-WAN overlay described in §3.6.2. This encrypts traffic in transit between the two sites. Encryption in transit is a distinct requirement from Ghana data-residency (TR-007): TR-007 concerns where client financial data is stored, backed up, replicated, and recovered, not how it is protected while moving between sites, so the IPsec mechanism described here does not by itself satisfy TR-007: data-residency compliance is addressed through the storage, backup, and recovery location decisions made in Sections 4 and 7.

3.6.5 Performance and Capacity

WAN transport is sized against TR-010's established performance targets. Primary Internet/WAN capacity is 1 Gbps symmetric at HQ and 500 Mbps symmetric at the Branch. Secondary transport capacity is sized to at least 50% of the corresponding primary link (500 Mbps at HQ, 250 Mbps at the Branch), consistent with TR-010's requirement that secondary connectivity maintain priority business services if the primary connection fails. Applying TR-010's 70% sustained-utilisation threshold to these WAN links specifically yields a planning target of 700 Mbps sustained / 300 Mbps headroom at HQ, and 350 Mbps sustained / 150 Mbps headroom at the Branch. These figures are carried forward from TR-010 without alteration; no new capacity figure is introduced in this section.

3.6.6 Routing and Failover

In production, path monitoring, selection and failover between the primary and secondary transports are primarily functions of the SD-WAN architecture. The exact carrier-edge routing and handoff arrangement, including whether BGP is required, depends on the selected providers and final service design. This is distinct from the Packet Tracer implementation, where inter-site routing is provided by a GRE tunnel carrying a static summary route (10.20.0.0/23) with static default routing at each router, as established with direct configuration evidence in §3.5.4. The Packet Tracer arrangement is a disclosed, intentional simulation representation of the inter-site relationship: it demonstrates the logical connectivity and routing behaviour called for by the Part B tasks, not the production SD-WAN/IPsec architecture itself, and is not presented as equivalent to it.

3.6.7 Final WAN Recommendation

GlobalBridge's production WAN and Internet connectivity strategy is dual-transport Secure SD-WAN (dedicated fibre primary, diverse secondary path, using different providers and physically diverse last-mile routing where feasible, at both HQ and the Branch) with site-to-site IPsec protecting inter-site traffic, sized to the capacity and utilisation targets in TR-010, and providing the redundancy required by TR-004/TR-005 through transport diversity rather than dependence on any single provider's advertised SLA. Final carrier selection remains subject to commercial evaluation and the WAN cost allowance established in Section 8, which prices the four required circuits but does not select a specific carrier: no Ghanaian carrier publishes enterprise circuit-level pricing, so RFQ engagement with candidate providers is required before procurement.

Section 4: Infrastructure and Equipment Selection

4.1 Network Equipment Selection & Justification

This section presents GlobalBridge's production network equipment selections, using a structured comparison for each of the eleven required equipment categories. Consistent with A-001, these are production recommendations, independent of the equipment families used inside the Packet Tracer simulation for the Part B configuration tasks.

4.1.1 Core Layer Switches (HQ)

Criterion	Requirement	Option A: Cisco Catalyst 9500	Option B: Aruba CX 8360	Option C	Selected?
Throughput / switching capacity	≥10 Gbps aggregation (TR-010)	Up to 6.4 Tbps on Catalyst 9500 models; up to 12.8 Tbps full duplex on Catalyst 9500X models	Substantially exceeds TR-010's requirement	N/A	Both exceed requirement
Redundancy/ HA	No single point of failure (TR-005)	StackWise Virtual, SSO, redundant power	Comparable HA capability	N/A	Both adequate
Operational fit	Consistency with rest of estate	Establishes single-vendor Cisco baseline for Distribution/ Access	Genuine vendor-diversity alternative (avoids Cisco-by-default)	N/A	Cisco preferred
Selected	—	Catalyst 9500 Series, deployed as a redundant pair	Not selected	—	Option A

Both families comfortably exceed the ≥10 Gbps aggregation requirement (TR-010), so raw capacity was not the deciding factor. The choice came down to enterprise-Core positioning, StackWise Virtual/HA capability, and single-vendor operational consistency with the Distribution and Access tiers below. This is a disclosed trade-off: Aruba's vendor-diversity advantage is genuine and not dismissed. A redundant Core pair is required to satisfy TR-005; a single Catalyst 9500, however capable, would remain a single point of failure. Exact SKU, quantity beyond the redundant pair, and pricing are deferred to Section 8.

4.1.2 Distribution Layer Switches (HQ)

Criterion	Requirement	Option A: Cisco Catalyst 9300	Option B: Aruba CX 6300	Option C	Selected?
Throughput / switching capacity	≥10 Gbps aggregation (TR-010)	Exceeds requirement	Exceeds requirement	N/A	Both adequate
Operational fit	Consistency with selected Core	Single-vendor consistency with Catalyst 9500 Core	Vendor-diversity alternative	N/A	Cisco preferred
Selected	—	Catalyst 9300 Series (family level)	Not selected	—	Option A

Selected on the same operational-consistency basis as the Core. Quantity (2× Catalyst 9300-24T-A, held at the minimum redundant-HA-pair floor rather than a port-count formula) and exact SKU are Section 8, Planning (C-003) items: final quantity may increase once HQ floor/MDF/IDF topology is known.

4.1.3 Access Layer Switches (HQ)

Criterion	Requirement	Option A: Cisco Catalyst 9200-48P	Option B: Cisco Catalyst 9300	Option C: Aruba CX 6300	Selected?
Wired user access	1 GbE, PoE+	Yes	Yes	Yes	All meet requirement
Uplink capability	≥10 Gbps aggregation (TR-010)	Modular 10G uplinks	Modular 10/25G uplinks	Exceeds requirement	All meet requirement
Stacking bandwidth	Not a required capability	StackWise-160	StackWise-480 (higher)	Comparable	Not deciding factor
SD-Access capability	Not required by this design	Supported, with model-dependent limits (C9200 / C9200L / enhanced-VN SKUs vary)	Supported with broader capabilities	N/A	Not a deciding factor
List price (48-port PoE+, estimate)	—	≈\$6,000	≈\$14,500	Not costed	Cost-decisive

Criterion	Requirement	Option A: Cisco Catalyst 9200-48P	Option B: Cisco Catalyst 9300	Option C: Aruba CX 6300	Selected?
Selected	—	Catalyst 9200-48P	Superseded (see below)	Not selected	Option A

Section 8's value-engineering review compared the Catalyst 9200 directly against the Catalyst 9300 for HQ Access and selected the Catalyst 9200-48P: it satisfies every established HQ Access requirement (1GbE, PoE+, $\geq 10\text{Gbps}$ uplink per TR-010) without paying for the higher stacking capacity and additional enterprise capabilities of the Catalyst 9300 that are not required by the HQ Access design, and it establishes a single common access-layer platform family with the Branch (§4.1.4 below). Quantity (20×, VLSM-derived from the established 800-host 3-year HQ projection excluding VOIP) is a Section 8 Planning (C-003) item and is not restated in full here.

4.1.4 Branch Core Switch

Criterion	Requirement	Option A: Cisco Catalyst 9200L	Option B: Aruba CX 6200	Option C	Selected?
Routing capability	Single-area OSPF (§3.5 Area 0 design)	Supported subject to selection of the required Cisco IOS XE feature/license level; final SKU/licensing must support the single-area OSPF design in §3.5	Single-area-OSPF-only capability: not a disadvantage given this project's Area 0 scope	N/A	Both adequate, subject to licensing
Scale fit	Sized to Branch (250 users, 3 VLANs), not inherited from HQ Core	Right-sized	Right-sized	N/A	Both adequate
Operational fit	Consistency with HQ Cisco estate	Single-vendor consistency	Vendor-diversity alternative	N/A	Cisco preferred
Selected	—	Catalyst 9200L (family level), 2× redundant pair	Not selected	—	Option A

Sized independently to the Branch's actual scale rather than automatically inheriting the HQ Catalyst 9500 selection. Aruba CX 6200 is confirmed a technically credible alternative; Cisco is preferred for operational consistency with the already-selected HQ switching environment. Quantity (2×, redundant pair, 2U total) is established at family/architecture level in §4.4: this is one of the few Branch categories with a quantity fixed in Section 4 itself, rather than deferred to Section 8's Planning framework.

4.1.5 Branch Access Switches

Criterion	Requirement	Option A: Cisco Catalyst 9200 (L/PXG mix)	Option B: Aruba CX 6300	Option C	Selected?
Wired user access	1 GbE, PoE+	Yes	Yes	N/A	Both meet requirement
Uplink capability	≥10 Gbps aggregation	Yes	Yes	N/A	Both meet requirement
Multigigabit option	Not auto-required by AP-635's 2.5GbE uplinks	Available as PXG variant, optional	Available, optional	N/A	Deferred to port sizing
Operational fit	Consistency with HQ/Branch Core platform	Single common family with HQ Access	Vendor-diversity alternative	N/A	Cisco preferred
Selected	—	Catalyst 9200 family	Not selected	—	Option A

The exact 9200L-vs-PXG mix is deferred to AP/port sizing: the AP-635's 2.5GbE uplink capability is explicitly not treated as an automatic wired-access requirement. Quantity (9×, using the same VLSM-based method as HQ Access, applied to the Branch's established 350-host 3-year total) is a Section 8 Planning (C-003) item.

4.1.6 WAN/Edge Routers (HQ and Branch)

Criterion	Requirement	Option A: Standalone edge router	Option B: Integrated SD-WAN/N GFW edge platform (§4.1.7/4.1.8)	Option C	Selected?
WAN termination, routing, path selection, failover, IPsec	Required at both sites	Requires separate router + firewall	Performed natively by the selected NGFW platform	N/A	Integrated platform sufficient

Criterion	Requirement	Option A: Standalone edge router	Option B: Integrated SD-WAN/N GFW edge platform (\$4.1.7/4.1.8)	Option C	Selected?
Architectural simplicity	Not a stated requirement, but not to be added without need	Adds a device and failure domain not currently justified	No additional device required	N/A	Integrated platform preferred
Selected	—	Not selected	No standalone production router at either site: function performed by the integrated SD-WAN/N GFW platform (\$4.1.7/4.1.8)	—	Option B

This category is included as a named comparison; the conclusion is that no standalone router is introduced at HQ or the Branch. This is explicitly conditional: a dedicated edge router would be introduced, without disturbing the underlying WAN architecture, if future ISP-handoff, BGP, or provider-managed-service requirements demonstrate a technical need. The Packet Tracer simulation's Cisco ISR4331 routers remain simulation-only devices, not a production recommendation, per A-001.

4.1.7 Next-Generation Firewall (NGFW): HQ Perimeter

Criterion	Requirement	Option A: Cisco Meraki MX250	Option B: FortiGate 100F	Option C: Meraki MX84	Selected?
Site-to-site VPN throughput	Must exceed HQ WAN capacity (1 Gbps primary)	4 Gbps maximum site-to-site VPN	1 Gbps: equals, not exceeds, HQ's primary capacity	250 Mbps: inadequate	MX250 clears with headroom
NAT firewall throughput	Adequate for HQ scale	7.5 Gbps maximum NAT firewall	Adequate at HQ scale	500 Mbps: inadequate at HQ population	MX250 selected
NGFW throughput	Adequate for HQ scale	2 Gbps maximum NGFW	Adequate at HQ scale	Inadequate at HQ population	MX250 selected

Criterion	Requirement	Option A: Cisco Meraki MX250	Option B: FortiGate 100F	Option C: Meraki MX84	Selected?
Device-count ceiling	Adequate for HQ population/scale	2,000 recommended devices	Adequate	~200-device recommended ceiling: inadequate	MX84 disqualified
Selected	—	Meraki MX250, HA pair	Not selected for HQ	Not selected	Option A

FortiGate 100F is not disqualified as a platform generally: its 1 Gbps IPsec VPN throughput figure equals, rather than exceeds, HQ's 1 Gbps primary WAN capacity, leaving no nominal headroom at HQ specifically, so a higher-capacity FortiGate model remains a credible alternative if pursued during procurement. The MX84 is rejected on its 500 Mbps firewall / 250 Mbps VPN throughput and ~200-device ceiling against HQ's population and 500 Mbps+ requirement. An HA security-gateway pair, not a single appliance, is required at HQ per TR-005. The MX250 figures above are drawn from the official [Cisco Meraki MX250 datasheet](#). The selected MX250's capacity figures should be verified against the current Cisco Meraki data sheet at procurement, together with the required security and SD-WAN licensing tier. The FortiGate 100F figure retains its earlier source qualification (a reseller blog, not a manufacturer-verified specification) in the final bibliography, since it was not selected and its figures were not re-verified against Fortinet's own current data sheet.

4.1.8 Branch Firewall/UTM

Criterion	Requirement	Option A: Cisco Meraki MX85	Option B: FortiGate 60F	Option C	Selected?
Site-to-site VPN/NGFW throughput	Must exceed Branch WAN capacity (500 Mbps primary)	1 Gbps-class firewall/NGFW/VPN performance ; meets the 500 Mbps Branch primary-WAN requirement, subject to final enabled-feature sizing	6.5 Gbps IPsec / 700 Mbps threat-protection: higher headroom	N/A	Both adequate

Criterion	Requirement	Option A: Cisco Meraki MX85	Option B: FortiGate 60F	Option C	Selected?
Device-count ceiling	Adequate for ~250 employees + phones / APs / infrastructure	250 devices recommended (official Meraki sizing guidance): close to Branch population, flagged for validation since 250 employees is not the same as 250 connected devices	Not the limiting factor at this throughput class	N/A	MX85 preferred, with a caveat
Operational fit	Consistency with HQ Meraki management	Single-vendor or consistency with HQ NGFW	Vendor-diversity alternative	N/A	Meraki preferred
Selected	—	Meraki MX85: preferred baseline, not final	Confirmed credible, higher-headroom alternative	—	Option A, conditionally

Sized independently against the Branch's actual WAN/employee numbers (500 Mbps primary/ $\geq$ 250 Mbps secondary, ~250 employees plus phones, APs, and infrastructure) rather than defaulting to the HQ MX250 selection, correcting the earlier MX84-at-Branch undersizing problem without repeating it as an oversizing problem via the MX250. This is explicitly flagged as not fully final: the MX85's ~250-device recommended scale is close enough to the Branch's actual population that final suitability must be validated against total connected-device count, not employee headcount alone, before procurement. If that calculation exceeds the MX85's operating scale, the design moves upward within the Meraki family rather than deploying an undersized appliance. Per the official [Cisco Meraki MX85 datasheet](#), each MX85 draws 55 W maximum, giving a 110 W planning maximum for the two-appliance HA pair; this is stated explicitly per-appliance so 110 W is not misread as the specification of a single unit. Quantity (2 $\times$, HA pair, deliberately split across A/B protected power paths since each appliance has a non-redundant internal PSU) is established at family/architecture level in §4.4.

4.1.9 Wireless Access Points

Criterion	Requirement	Option A: Aruba AP-635	Option B: Cisco Catalyst 9130AX	Option C	Selected?
Spectrum flexibility	Not a fixed requirement, evaluated as a genuine trade-off	Tri-band 2.4/5/6 GHz, Wi-Fi 6E	2.4/5 GHz only	N/A	AP-635 advantage
Spatial-stream capability	Not a fixed requirement, evaluated as a genuine trade-off	2×2:2 per band	8×8 5 GHz MU-MIMO	N/A	9130AX advantage
Uplink	Not a fixed requirement	Dual 2.5GbE uplinks	Standard uplink	N/A	AP-635 advantage
Selected	—	Aruba AP-635	Not selected	—	Option A

Neither platform is presented as universally superior: the Catalyst 9130AX wins on spatial-stream capability, and the AP-635 wins on spectrum flexibility and uplink capacity. The trade-off is stated explicitly rather than flattened into a one-sided win. AP quantity and placement are explicitly not calculated from headcount: per C-003 (no floor plans, construction materials, RF interference data, or cable distances supplied), a predictive design followed by an RF site survey governs final count and placement, not a formula from the employee total. The Section 8 planning figure (32×, derived from Aruba’s own RF design guidance applied against the case study’s device-growth ratio) is a capacity-based budgeting estimate, not a coverage/floor-plan-validated count.

4.1.10 Wireless LAN Controller (WLC) / Cloud Management Platform

Criterion	Requirement	Option A: Cloud-managed (Aruba Central)	Option B: Standalone hardware WLC	Option C	Selected?
Management architecture	Consistent with selected AP-635 platform	Native cloud management, no on-prem controller required	Adds a dedicated on-prem appliance and failure domain not required by the selected AP platform	N/A	Cloud-managed preferred
Selected	—	Aruba Central, cloud-managed architecture	Not selected: standalone WLC not required	—	Option A

A standalone hardware WLC is not required under the selected cloud-managed architecture. Aruba Central licensing, cloud-service region, data handling, and retention must be reviewed before procurement. Where any information processed by the service falls within the client-financial-data scope of TR-007, its storage, backup, replication, and recovery locations must comply with the Ghana residency requirement; TR-007 is not read as automatically extending to every network-management log or AP telemetry record, consistent with the same security-control/TR-007-scope distinction already drawn in §3.6. This is stated as a genuine open review item, not a silently assumed pass.

4.1.11 UPS Systems (HQ Data Centre and Branch)

Criterion	Requirement	Option A: Eaton 9PX G2	Option B: APC Smart-UPS On-Line SRT	Option C	Selected?
Topology	Online double-conversion	Yes	Yes	N/A	Both meet requirement
Capacity range	Must cover final connected load (not yet fully evidenced)	3–11 kVA	Extends to 20 kVA	N/A	Neither selectable yet on evidence
Selected	—	Preferred, provisional	Confirmed credible alternative	—	Provisional, not final

UPS Systems (HQ Data Centre and Branch) is a single equipment category, and this comparison is presented once accordingly: per-site kVA sizing is a quantity/capacity matter, not a separate structural category. Topology (online double-conversion) is fixed; the specific family, kVA rating, runtime, and quantity remain provisional because they depend on actual connected server/storage/network equipment load, which §4.2–§4.4’s power-evidence work has only partially established (867.4W HQ confirmed subset; 110W Branch). Specifying a particular model or kVA rating without that evidence would not be supportable. Section 8’s UPS line (Eaton 9PX 6kVA×2 HQ, 3kVA×2 Branch) is an explicitly disclosed budgetary planning allowance, not final engineering sizing.

4.1.12 Dedicated Storage-Network Switching (Fabric A/B)

Not one of the eleven named §4.1 equipment categories; included here because it is a §4.1-level family/architecture switching selection, resolved during the Section 4 Cross-Section Reconciliation Audit, and belongs alongside the other switching decisions rather than only inside §4.3’s storage narrative.

Criterion	Requirement	Option A: Cisco Catalyst 9300X-24Y	Option B: Cisco Nexus 93240YC-F X2	Option C	Selected?
Native 25GbE SFP28 ports	Required for iSCSI host/SAN connectivity	24 ports	48 ports	N/A	Both meet requirement
Fabric A/B with two switches	Required for dedicated redundant fabric	Yes	Yes	N/A	Both meet requirement
Maximum power consumption	Not a fixed requirement, cost-relevant	243.7 W	708 W	N/A	9300X-24Y advantage
Operational model	Consistency with campus estate	IOS-XE, consistent with campus	NX-OS, additional operational platform	N/A	9300X-24Y advantage
Scale fit	Sized to 2 Hyper-V hosts + 1 dual-controller SAN	Strong fit	Considerably more capacity than currently justified	N/A	9300X-24Y sufficient
Selected	—	Catalyst 9300X-24Y, 2× (Fabric A/Fabric B)	Not selected	—	Option A

This was an independent comparison of two suitable storage-switch platforms, not an automatic reuse of the general “Catalyst 9300 Series” campus selection: the base Catalyst 9300 family does not itself carry native 25GbE capability, which sits specifically on the 9300X sub-family, selected here on its own merits (required native 25GbE, sufficient scale, substantially lower maximum power, and IOS-XE operational consistency). Production shared storage uses these two dedicated redundant switches, physically and logically separate from the general campus LAN, rather than carrying iSCSI traffic over the Distribution/Access fabric via VLANs, preserving the clean failure-domain separation that §4.3’s SAN architecture depends on. Storage-network path redundancy (STOR-NET-001): each Hyper-V host and the dual-controller PowerVault ME5024 have redundant connectivity across both fabrics, with Microsoft MPIO providing host multipathing. Exact port assignments, VLAN/subnet configuration, MTU, flow control, and transceiver compatibility remain implementation-validation items, not assumed from platform capability alone.

4.1.13 Production versus Packet Tracer Equipment

Consistent with A-001, the equipment families selected above are production recommendations, evaluated on current data-sheet evidence and this project’s own requirements. They are independent of the Packet Tracer simulation’s device inventory, which supports the Part B configuration tasks within Packet Tracer’s own device-model constraints.

Production role	Production recommendation	Packet Tracer simulation role
Core	Redundant Catalyst 9500 pair	3650/3560-class simulation devices
Distribution / Access (HQ, Branch)	Catalyst 9300 (Distribution) / Catalyst 9200 (Access, HQ and Branch)	2960-class simulation devices
WAN/Edge	Integrated SD-WAN/NGFW (Meraki MX250 HQ, MX85 Branch): no standalone router	Cisco ISR4331

Cisco’s own end-of-life notice for the Catalyst 2960 family used in the simulation (end-of-sale 31 October 2014; security support ended 30 October 2017; end-of-support 31 October 2019) is the disclosed reason 2960-class equipment cannot be a production recommendation, regardless of its adequacy inside the simulation.

4.2 Server Infrastructure

This section specifies GlobalBridge’s production server infrastructure against the nine required server roles. For each role, the fixed platform and quantity constraint is applied first, before any team-determined CPU, RAM, storage, or operating-system value.

4.2.1 Workload and Virtualization Design Approach

The nine required server roles are designed to their fixed platform and quantity constraints first. Virtual-designated workloads are consolidated onto a resilient hypervisor cluster, while physical-designated workloads (the ERP database and the Backup & Recovery server) remain outside it, per the platform specification for those two roles. Hypervisor capacity is calculated from actual VM resource demand plus a declared engineering reserve, three-year growth treatment, and N+1 spare capacity, rather than a fixed vCPU:pCPU consolidation ratio: a static ratio is explicitly rejected as outdated, consistent with current virtualization guidance. Every CPU, RAM, and storage figure in this section is labeled as one of three provenance categories: source-derived requirement, calculated design value, or declared planning assumption. No figure is presented as fact without that label being traceable.

Microsoft Hyper-V with Windows Server Failover Clustering is the selected production hypervisor platform, chosen on proportionality, integration, and certification grounds rather than cost. The workload is a compact, Windows-environment-integrated estate, not a scale or feature profile that demonstrates a need for VMware vSphere’s broader clustering and automation feature set. Proxmox VE was not selected because its hardware-certification evidence for the candidate Dell/HPE server models remains unresolved. This selection is not cost-driven; it remains subject to revision if Section 8’s TCO analysis demonstrates a materially poorer outcome than a technically compliant alternative, and no such outcome has been identified in that analysis.

4.2.2 Consolidated Nine-Role Server Specification Table

No.	Server Role	Platform	CPU	RAM	Storage	NIC / Connectivity	OS	Qty
1	Active Directory / DNS	VM	2 vCPU	8 GB	60 GB system disk (engineering allocation)	1 production vNIC/VM (redundancy from host-level 2×25GbE, not multiple vNICs)	Windows Server 2025 (architectural consequence of the selected Microsoft AD DS/DNS design, not a universal directory/DNS requirement)	2 (High Availability, separate dedicated cross hosts)
2	DHCP	VM	2 vCPU	4 GB	40 GB system disk (engineering allocation)	1 production vNIC/VM	Windows Server 2025 (integrates with the selected Microsoft directory/DNS environment)	1
3	File & Print	VM (clustered)	2 vCPU	4 GB	60 GB system disk/VM + 20.79 TB logical shared data (3-year; delivered on the shared SAN, §4.3)	1 production vNIC/VM	Windows Server 2025 (AD identity / permissions / SMB integration)	2 (clustered)
4	Web / Intranet	VM (load-balanced)	2 vCPU	4 GB	60 GB system disk (engineering allocation)	1 production vNIC/VM; load-balancing mechanism Decision Required (not assigned to any specific product)	Ubuntu Server 24.04 LTS (no established Windows-specific application requirement; avoids adding Windows guests/licensing for a generic lightweight portal)	2 (load-balanced)

No.	Server Role	Platform	CPU	RAM	Storage	NIC / Connectivity	OS	Qty
5	Email (M365 hybrid)	Cloud (Microsoft 365): evaluated/preferred operationally; production selection conditional on TR-007 residency confirmation	N/A (cloud)	N/A (cloud)	N/A (cloud)	N/A (cloud)	N/A: no local server OS under the evaluated cloud architecture	N / A
6	Database / ERP	Physical (2U, Dell PowerEdge R760 or HPE ProLiant DL380 Gen11: vendor Decision Required, §4.1/4.2)	8 physical cores	128 GB	≥2.16 TB usable DB capacity/server (1.728 TB 3-yr working DB at an 80% utilization ceiling) + declared transaction-log and tempdb allowances, on a shared RAID10 SSD array/server (§4.3); RAID controller model Decision Required	Redundant ≥10GbE production; separate replication NIC recommended, not mandatory	Decision Required: dependent on the selected ERP/database platform (no ERP product is specified for this project, and ERP was deliberately sized as product-neutral rather than naming a platform solely to complete this cell)	2 (primary + replica)

No.	Server Role	Platform	CPU	RAM	Storage	NIC / Connectivity	OS	Qty
7	Backup & Recovery	Physical, 2U LFF-capable rack server (Dell PowerEdge R760 / HPE ProLiant DL380 Gen11 class: required by the 11-drive internal repository; the 1U R660/DL360 class used for the hypervisor hosts cannot accommodate it)	8 physical cores (project sizing baseline for backup data movement, repository operations, restore processing, and tape interaction: independently derived, not reused from the ERP server's coincidentally identical figure)	64 GB (informed by the repository's ReFS memory consideration plus OS / Veeam / operational overhead; not a universal Veeam requirement)	11 × 8 TB enterprise NL-SAS (10-member RAID 6 + 1 hot spare) internal repository, ≥64 TB theoretical protected capacity against a ≥51 TB planning requirement; ReFS 64 KB allocation unit; 2× mirrored enterprise M.2 boot; separate LTO-9 tape tier via a preferred HPE StoreEver MSL3040 (3U, 40-slot, 1×LTO-9 drive initially), 32-cartridge minimum calculated GFS retention-media baseline (a planning baseline, not a procurement quantity)	Redundant ≥10GbE (upgraded from single ≥10GbE once the project-wide A/B electrical path architecture made a single-NIC dependency inappropriate)	Windows Server 2025 (traditional Veeam Backup & Replication deployment, the project's reference backup architecture)	1

No.	Server Role	Platform	CPU	RAM	Storage	NIC / Connectivity	OS	Qty
8	SIEM / Log Management	6 VMs, distributed Wazuh architecture: 1 master + 2 workers + 3 indexers (one indexer co-located with the dashboard, per Wazuh's own permitted architecture)	Master: 8 vCPU; 2×Worker: 8 vCPU each; 2×Indexer: 8 vCPU each; 1×Indexer + Dashboard: 12 vCPU: 52 vCPU total (Wazuh-anchored planning allocation, platform-dependent, not final)	Master: 4 GB; 2×Worker: 4 GB each; 2×Indexer: 16 GB each; 1×Indexer + Dashboard: 24 GB: 68 GB total	1.2672 TB ingest-derived baseline (14.08 GB/day × 90 days) × 2 (one index replica) = 2.5344 TB, ÷0.80 utilization ceiling → ≥3.2 TB usable SIEM flash-tier storage (RAID10 SSD, a separate disk group on the shared SAN, §4.3); system disks 60 GB × 6 VMs = 360 GB (project allocation)	1 production vNIC/VM (×6); no dedicated load-balancer VM (a load-balancer is recommended, not required, per Wazuh's own multi-server-address agent failover)	Ubuntu Server 24.04 LTS (×6) (Wazuh's reference architecture is Linux-based)	6

No.	Server Role	Platform	CPU	RAM	Storage	NIC / Connectivity	OS	Qty
9	Hypervisor Host(s)	Physical, 1U rack (Dell PowerEdge R660 or HPE ProLiant DL360 Gen11, vendor Decision Required)	1× Intel Xeon Gold 5418Y, 24 physical cores (single socket populated of two available); PASS subject to workload validation, not an unconditional capacity proof	192 GB (corrected from an earlier 128 GB baseline after N+1 re-validation against the revised 13-VM workload; 192–130=62 GB remains outside the reserved VM envelope under one-host failure)	Mirrored M.2 boot only: no local VM data storage; shared/cluster storage is delivered via the dual-controller external SAN (§4.3), not Storage Spaces Direct/Azure Local	Redundant 2×25GbE	Windows Server 2025, Server Core, with the Hyper-V role (the free standalone Hyper-V Server SKU no longer exists as of Windows Server 2025, so the host OS itself requires a licensed Windows Server edition)	2

4.2.3 SIEM Architecture: Distributed Wazuh Topology

The SIEM/Log Management role is implemented as a 6-VM distributed Wazuh deployment rather than the single VM originally scoped, a correction made after evaluating Wazuh’s own architectural constraints. Wazuh permits only one server-cluster master, an architectural limit rather than a configurable choice. Two server workers preserve log collection, decoding, rule evaluation, and alert generation if the master or one worker becomes unavailable. Three indexer nodes form a quorum-oriented searchable-index tier, selected on Wazuh-maintainer guidance favouring three nodes over two to reduce split-brain risk. The dashboard is co-located on one indexer VM, which Wazuh explicitly permits, rather than assumed to reside on the master.

The Wazuh master remains a documented residual single point of failure. Wazuh does not provide native automatic master failover, so its loss primarily affects SIEM administration, agent registration,

configuration synchronisation, API access, and dashboard availability, not log collection, decoding, or alerting, which continue on the surviving worker nodes. This limitation is accepted and disclosed rather than concealed; the master VM is protected by the underlying Hyper-V host-failover architecture and SAN resilience, backed up regularly, and supported by a documented restoration/rebuild runbook. Host-level N+1 failover restarts the master VM on a surviving host after a physical-host failure, but this does not create a second Wazuh master and does not protect against application- or OS-level corruption. The SIEM as a whole is therefore not described as fully highly available without this qualification: collection and detection are resilient (two or more workers), while the management/control plane is not.

The index replica policy is set to one replica per index (Wazuh's own default ships with zero replicas, which the production design deliberately changes), allowing the full index to remain available after one indexer-node failure, consistent with TR-005's single-point-of-failure removal requirement. A second replica is not adopted, since TR-005 requires removing a single-node dependency, not surviving two simultaneous failures, and a second replica would increase flash capacity requirements without a demonstrated need.

4.2.4 Hypervisor Host Sizing and N+1 Validation

The two-host N+1 design is validated by a deterministic RAM test: the surviving host's capacity after one host failure must meet or exceed the full three-year workload demand. Against the corrected 13-VM estate (66 vCPU / 104 GB baseline, $\times 1.25$ declared engineering reserve $\rightarrow$ 82.5 vCPU / 130 GB three-year planning demand), the originally specified 128 GB per host fails this test ($128 < 130$) once the SIEM topology correction is applied. Host RAM is accordingly revised to 192 GB per host, leaving 62 GB of physical RAM outside the reserved VM envelope under one-host failure, available for hypervisor overhead and operational margin.

CPU is deliberately not tested via any vCPU-to-physical-core conversion ratio: this project has consistently rejected inventing such a ratio, since allocated vCPU counts do not establish simultaneous 100% utilisation across every VM, and no defensible conversion exists in either Wazuh's or this project's evidence. The retained 24-physical-core-per-host configuration is stated as "PASS subject to workload validation," not an unconditional capacity proof; production monitoring of physical CPU utilisation, Hyper-V processor-queue-equivalent indicators, and Wazuh's own analysis/remoted dropped-event counters is the established validation path instead.

4.2.5 ERP Database Servers

The ERP database role uses a fixed physical architecture: two dedicated 2U rack servers, primary and replica, each independently sized to 8 physical cores and 128 GB RAM as a declared, externally sanity-checked planning baseline, not tied to any named ERP product, since none is specified for this project. Storage is sized from a declared 1.0 TB initial working database footprint per instance, grown at a declared 20% compound annual allowance to a 1.728 TB three-year working footprint, then divided by an 80% maximum planning-utilisation ceiling to reach ≥ 2.16 TB usable database-storage capacity per server, before separate allowance for transaction logs, tempdb, and OS/platform overhead. The replica independently requires the same 2.16 TB, a consequence of holding a complete copy, not a separate assumption, but the ERP dataset is counted once (1.728 TB) in the protected-data backup inventory, since the replica is availability infrastructure, not a second protected dataset.

The one item this project deliberately leaves open throughout §4.2 is the ERP database platform and its operating system: no ERP product is specified for this project, and Section 4.2's sizing was kept product-neutral by design rather than resolved by naming a platform merely to complete a table cell. This is disclosed as a transparent dependency, not a gap in the design reasoning.

4.2.6 Backup & Recovery Server

The Backup & Recovery server is one of the required physical server roles, sized from the fully resolved backup capacity chain: a 22.918 TB conventional protected-data baseline (File & Print, ERP, and ordinary VM system disks; the SIEM searchable dataset is explicitly excluded, protected instead by the SIEM platform's own retention), a 14-day local operational recovery window at declared per-component daily change rates, and an 80% utilisation ceiling, yielding a ≥ 51 TB usable local-repository planning target. The server's own CPU (8 physical cores) and RAM (64 GB) are project-selected planning baselines, informed respectively by the backup role's data-movement/repository/restore/tape workload and by the repository's ReFS memory consideration plus operational overhead. Neither figure is presented as a universal Veeam requirement, and final utilisation must be validated under representative backup, synthetic-full, and restore workloads.

Longer-term retention is provided by an independent-full-generation LTO-9 tape tier (4 weekly + 12 monthly GFS generations, calculated at a minimum of 32 LTO-9 cartridges from the 22.918 TB baseline at 18 TB native capacity per cartridge), stored offline and at a geographically separate Ghana location consistent with TR-007. The 32-cartridge figure is stated explicitly as a minimum calculated retention-media baseline, not a procurement quantity: actual media requirements may exceed it for scratch media, cleaning cartridges, rotation overlap, and platform-specific tape-job behaviour.

4.2.7 Virtual Machine Estate Summary and Operating System Mix

The complete virtual estate totals 13 VMs (7 non-SIEM VMs unchanged from the original design plus the 6-VM Wazuh topology, replacing the original single SIEM VM): 66 vCPU and 104 GB RAM as the workload baseline, rising to 82.5 vCPU and 130 GB under the declared 25% engineering reserve applied for N+1 validation. 760 GB of ordinary VM system disk is required (400 GB across the seven non-SIEM roles, 360 GB across the six Wazuh VMs), delivered as ≥ 1 TB usable RAID10 enterprise-SSD VM datastore capacity (§4.3) after the same utilisation ceiling.

Operating-system assignment follows the architecture already selected, not intuition. AD/DNS, DHCP, File & Print, and the Backup & Recovery server run Windows Server 2025 as a consequence of the selected Microsoft Active Directory/backup architecture, not a universal requirement. Web/Intranet and all six Wazuh VMs run Ubuntu Server 24.04 LTS, since neither role has an established Windows-specific requirement and Wazuh's own reference architecture is Linux-based. The hypervisor hosts run Windows Server 2025 Server Core with the Hyper-V role, a direct consequence of the Hyper-V platform decision (the free standalone Hyper-V Server SKU no longer exists as of Windows Server 2025). This yields 5 Windows Server guest VMs and 8 Linux guest VMs within the 13-VM estate. Section 8's Windows Server licensing calculation separately accounts for the selected Hyper-V host licensing and the Windows Server guest rights required by this architecture (Standard edition selected over Datacenter at this VM density, closed in Section 8).

4.2.8 Virtual NIC Policy

VM-level virtual NIC count is distinguished from physical hypervisor host NIC count (2×25GbE per host, redundant). Ordinary infrastructure VMs receive one production vNIC each, unless a specific technical requirement for additional logical network separation is identified: redundancy for these VMs comes from the underlying virtual switching and the hosts' redundant physical network paths, not from provisioning multiple vNICs per VM. The two physical ERP servers warrant separate consideration: redundant production connectivity plus a recommended, not mandatory, logically/physically separated replication path.

4.2.9 Data Centre Architecture Revalidation

Section 3.1's Core-attached (non-spine-leaf) data-centre architecture decision is revalidated against the corrected server estate (13 VMs across two hypervisor hosts, two physical ERP servers, one

physical backup server). No workload evidence demonstrates the server density or sustained east-west traffic pattern that would justify a dedicated spine-leaf fabric: the growth from 8 to 13 VMs is driven entirely by the Wazuh high-availability topology, not by new business workload, and does not change the underlying traffic pattern. The Core-attached architecture is retained, using redundant high-speed connectivity consistent with TR-005 and TR-010, subject to reconsideration if future server density, east-west traffic, or application scale materially exceeds the three-year design model.

4.3 Storage Architecture

This section specifies the physical storage architecture supporting the server infrastructure defined in §4.2. It covers the shared clustered-storage platform for the Hyper-V estate, dedicated physical storage for the ERP database servers, and the Backup & Recovery server's internal repository.

4.3.1 Storage Requirements Classification

Six storage domains were classified against capacity, workload type, performance priority, availability requirement, growth basis, backup relationship, and placement. This classification was derived strictly from the figures already established in §4.2, before any RAID level, drive technology, or storage-platform choice was made: the VM datastore (hypervisor system/application volumes), File & Print logical data, ERP primary/replica database storage, SIEM storage, the backup repository, and boot/system storage for the physical servers (already addressed via mirroring in each server's own specification). This exercise surfaced the central architectural question resolved in §4.3.2: compute-level N+1 alone does not satisfy TR-005 for the VM workloads if the surviving Hyper-V host cannot access the failed host's VM disks after a failure.

4.3.2 Shared Clustered Storage Architecture

A redundant external block-storage SAN was selected for the two-node Hyper-V cluster, evaluated against Windows Storage Spaces Direct (S2D), Azure Local, and NAS/SOFS using manufacturer-primary evidence. The production design uses a dual-controller array accessible by both hypervisor hosts through redundant high-speed storage-network paths with Microsoft MPIO, so no single controller, switch, interface, or storage link is the sole access path to clustered VM storage. Ethernet/iSCSI is preferred over Fibre Channel, since the existing server design already provides high-speed Ethernet connectivity and no requirement calls for a separate FC fabric.

S2D was evaluated as technically capable at this two-host scale but was not selected: it would require reopening the already-established hypervisor host design to add local data drives, it mandates Windows Server Datacenter edition, and it introduces two-node quorum and storage/compute convergence considerations without a demonstrated benefit at this project's 13-VM scale. With only two hosts, a single host failure already places the entire VM compute load on the survivor, and S2D would additionally leave the storage layer in a degraded two-node state during the same event. Azure Local was not selected either, since it adds licensing and Azure control-plane dependencies not required by the current design; its cloud-sync relationship was treated as an unrequired assurance dependency that counts against proportionality, though it was not established as a proven TR-007 violation, since a quorum witness holds cluster-arbitration information rather than protected client financial data. SOFS is treated as a storage-presentation mechanism rather than a separate underlying architecture and is not required.

Selecting SAN over S2D means the storage architecture does not force Windows Server Datacenter edition: Standard-vs-Datacenter remains a genuine Section 8 licensing decision against the 5-Windows-guest-VM figure, not one forced by the storage platform (Standard was ultimately selected, per Section 8). The established path-redundancy requirement is that each Hyper-V host has at least two independent high-speed storage paths, distributed across redundant storage-network switches and terminating across both controllers of the shared array, with Windows MPIO providing path failover and load distribution. As a result, no host, storage controller, storage switch, or

individual storage link constitutes the sole path between a hypervisor and the shared datastore. This is delivered via the dedicated redundant Fabric A/Fabric B storage-network switching selected in §4.1.12 (2× Catalyst 9300X-24Y), physically and logically separate from the general campus LAN.

The array carries three logically distinct workloads on shared infrastructure: the Hyper-V VM datastore, the SIEM searchable-index tier, and the File & Print bulk data. Using logically separate storage pools, volumes, and performance tiers per workload, rather than one undifferentiated LUN, is accepted only on condition that the array itself carries the dual-controller/redundant-path/MPIO protection described above.

4.3.3 Shared SAN Physical Layout

Storage domain	Physical layout	Protection	Raw capacity	Theoretical protected capacity	Usable target
VM datastore	4 × 1.92 TB enterprise SSD	RAID 10	7.68 TB	~3.84 TB	≥1 TB usable
SIEM index tier	4 × 1.92 TB enterprise SSD (separate disk group)	RAID 10	7.68 TB	~3.84 TB	≥3.2 TB usable
File & Print	6 × 8 TB enterprise NL-SAS HDD	RAID 6 (4 data + 2 parity)	48 TB	~32 TB	≥26 TB usable

All capacities are decimal (1 TB = 10¹² bytes); formatted/OS-visible capacity reads lower. Combined, the shared SAN carries 14 data drives (8 × 1.92 TB SSD + 6 × 8 TB NL-SAS HDD), 63.36 TB total nominal raw capacity, and 39.68 TB total theoretical protected capacity across the three groups. These are stated explicitly as theoretical post-RAID figures, not guaranteed formatted usable capacity, before array reserves, metadata, and formatting overhead.

The VM datastore and SIEM index tier are established as separate disk groups despite sharing the same SSD/RAID 10 configuration, keeping them logically distinct storage domains even where they share underlying media. The VM datastore’s 3.84 TB theoretical protected capacity substantially exceeds its ≥1 TB usable target. This margin results from the project’s selected four-drive RAID 10 layout using 1.92 TB enterprise SSDs, not from a requirement for 3.84 TB of VM capacity, and it provides additional expansion headroom. The SIEM tier carries a materially tighter margin (~0.64 TB above its ≥3.2 TB target), flagged as a procurement validation item: if validated usable capacity after array reservations and formatting overhead falls below 3.2 TB, SSD capacity is increased rather than the SIEM requirement reduced.

For File & Print, conventional RAID 6 (4 data + 2 parity) is selected over Dell’s ADAPT distributed-parity scheme: ADAPT’s 12-drive minimum is disproportionate to the ~26 TB requirement and consumes substantially more drive bays than the six-drive conventional layout, and no quantified capacity/rebuild advantage for ADAPT could be established from vendor-primary evidence for this specific configuration. ADAPT is not disqualified on technical merit, only not selected as the more transparent, proportionate choice here. Media is enterprise capacity HDD

appropriate to the selected array, with the exact drive class named only once the final array configuration is selected.

The array architecture is a single dual-controller SAN system, with SSD disk groups for VM/SIEM workloads and HDD capacity storage via a supported expansion enclosure attached to the SSD-based controller. Two principal pools are used: Pool A carries the VM datastore RAID 10 SSD disk group and the SIEM RAID 10 SSD disk group (both performance-oriented, independent RAID groups/volumes within the shared pool); Pool B carries the File & Print RAID 6 HDD disk group alone. Tier affinity/pinning is used so the design does not depend on automatic tiering moving data between SSD and HDD classes.

4.3.4 ERP Dedicated Physical Storage

Each of the two ERP database servers (§4.2) carries its own dedicated 4×1.92 TB enterprise SSD RAID 10 array, physically independent of the shared SAN, since ERP is a physical, not virtualized, workload with the highest random-IOPS, lowest-latency-tolerance profile in the estate.

Component	Capacity per ERP server
Database provisioned target	2.160 TB
Transaction log provisioned target	0.648 TB
tempdb project allowance	0.512 TB
Total planned allocation	3.320 TB
RAID 10 theoretical protected capacity	3.840 TB
Remaining theoretical margin	0.520 TB

The 2.16 TB database target is carried forward from the established §4.2 ERP storage assumption. The 648 GB transaction-log target is derived from a practitioner (not Microsoft) 30% starting allowance against the 1.728 TB three-year working database, then divided by the project's 80% planned-utilisation ceiling ($1.728 \times 0.30 = 0.5184$ TB, $\div 0.80 = 0.648$ TB). This is stated explicitly as practitioner guidance, not a Microsoft requirement, since Microsoft's own guidance is to size and monitor logs against actual log-generating operations rather than a fixed ratio. The 512 GB tempdb allowance is a declared project engineering assumption, carrying no borrowed vendor authority, since neither Microsoft nor independent practitioner sources provide a defensible tempdb-sizing ratio. The candidate 4×1.92 TB RAID 10 array passes two independent checks: the simple fit check (3.32 TB planned allocation against 3.84 TB protected capacity, 0.52 TB margin) and a stronger 80% utilisation sanity check using pre-ceiling working consumption ($1.728 + 0.5184 + 0.512 = 2.7584$ TB against $3.84 \times 0.80 = 3.072$ TB). This confirms the design remains beneath the project's utilisation threshold even before crediting the ceiling already applied to the database/log figures individually.

Database, transaction log, and tempdb use separate logical volumes on the same four-drive RAID 10 set, not separate physical RAID groups. This is a disclosed cost/performance trade-off, since separate physical groups for log/tempdb at this scale would consume substantially more drives without workload evidence proving physical separation is necessary; physical separation remains available if production performance testing later demonstrates contention. Both ERP servers receive identical storage ($2 \times 4 \times 1.92$ TB = 8×1.92 TB across the pair, 15.36 TB combined raw, 7.68 TB combined theoretical protected). Each server independently holds the full database capacity so the replica can assume the service role, but this is not a shared or halved 7.68 TB database, and the ERP dataset remains counted once (1.728 TB) in the backup protected-data inventory, not doubled because a replica exists.

4.3.5 Backup & Recovery Repository Physical Layout

The Backup & Recovery server's internal repository uses 10×8 TB enterprise NL-SAS drives in RAID 6 (8 data + 2 parity), plus 1×8 TB hot spare: 11 drives installed in total, using internal direct-attached storage in the dedicated physical server rather than an external storage array or shelf.

Attribute	Selected design
Repository media	Enterprise NL-SAS HDD
RAID members	10×8 TB
Protection	RAID 6, 8 data + 2 parity
Hot spare	1×8 TB
Total installed HDDs	11×8 TB
RAID-member raw capacity	80 TB
Total installed raw capacity (incl. spare)	88 TB
Theoretical protected capacity	64 TB
Minimum required capacity	≥ 51 TB usable
Theoretical margin before platform overhead	13 TB
Filesystem	ReFS, 64 KB allocation unit
Network	Redundant ≥ 10 GbE

Two larger candidates were evaluated and not selected. A 9×12 TB RAID 6 configuration was technically valid but carried substantially more unused protected capacity than needed, with larger RAID members that, all else equal, increase the data involved in rebuilding a single failed drive. A RAID 60 configuration was technically valid but would require 13 LFF bays, exceeding the Dell PowerEdge R760's 12-bay internal maximum and either constraining the vendor choice to HPE or requiring external storage on Dell. No requirement justifies that architectural penalty, since the repository is only one link in the established protection chain (production data, local RAID-protected repository, offline LTO-9, offsite Ghana-held media), and RAID 6 provides proportionate local disk-failure protection without needing the repository to be independently disaster-proof on its own. Eleven drives fit within the R760's 12-bay maximum (one bay remaining unoccupied), with more headroom on the HPE ProLiant DL380 Gen11 alternative. Internal storage was preferred over an external storage array or shelf because the repository does not require the shared-storage and live-migration capabilities of the Hyper-V SAN, and because housing production VM storage and its own backups on the same SAN platform would create a poor resilience design.

ReFS with a 64 KB allocation unit is selected for the Windows-based Veeam repository, since Veeam documents block-cloning/fast-clone integration benefits for applicable backup operations; capacity savings from ReFS fast-clone, compression, or deduplication are not credited toward the ≥ 51 TB physical usable-capacity requirement. ReFS carries an approximate 0.5 GB RAM per 1 TB filesystem-memory consideration, which at the 64 TB theoretical protected volume contributes roughly 32 GB. This is a component of, not the entirety of, the Backup & Recovery server's established 64 GB RAM specification (§4.2), which also accounts for Windows Server, Veeam services, and operational overhead.

The calculation ($10 \times 8 = 80$ TB RAID-member raw; $(10 - 2) \times 8 = 64$ TB theoretical protected against the ≥ 51 TB requirement, 13 TB margin) is explicitly not divided by the project's 80% utilisation ceiling a second time: the established ≥ 51 TB target already contains that margin, and a further deduction would double-count it. Final procurement must demonstrate actual usable-after-overhead capacity ≥ 51 TB as a validation item, not a further sizing calculation.

4.3.6 Consolidated Storage Architecture

Domain	Physical design	Protection	Capacity result
VM datastore	4 × 1.92 TB SSD	RAID 10	~3.84 TB theoretical
SIEM indexes	4 × 1.92 TB SSD	RAID 10	~3.84 TB theoretical, ≥ 3.2 TB required
File & Print	6 × 8 TB NL-SAS	RAID 6	~32 TB theoretical, ≥ 26 TB required
ERP Primary	4 × 1.92 TB SSD	RAID 10	~3.84 TB/server
ERP Replica	4 × 1.92 TB SSD	RAID 10	~3.84 TB/server
Backup repository	10 × 8 TB NL-SAS + 1 spare	RAID 6	~64 TB theoretical, ≥ 51 TB required
Offline/offsite backup	LTO-9	Independent full recovery points	32-cartridge calculated GFS minimum baseline

This is a coherent end-to-end design: a shared redundant SAN for clustered infrastructure (VM datastore, SIEM, File & Print, with dual-controller, MPIO, TR-005-compliant path redundancy delivered over the dedicated Fabric A/B storage switching in §4.1.12); dedicated transactional SSD storage for ERP, physically independent of the shared SAN; an independent capacity-oriented internal disk repository for backup, deliberately not sharing the production SAN to avoid a shared-failure-domain design flaw; and offline/offsite LTO-9 tape for recovery isolation consistent with the established 3-2-1-1-0 backup policy defined in Section 7.

Remaining items are explicitly procurement-level, not architectural: exact supported drives/controllers/array SKUs across all domains, final confirmation that the SIEM RAID 10 group remains ≥ 3.2 TB usable after array reservations, and Section 8 pricing/TCO for the complete storage bill of materials.

4.4 Data Centre Design

This section specifies rack, power, and environmental design for the equipment estate defined across §4.1–§4.3. It covers the remaining tape/Branch physical equipment data not already captured, the electrical power-path architecture, UPS sizing status, rack selection at both sites, and environmental design principles.

4.4.1 Tape Library, Branch Core, and Branch Security: Physical Data

The HPE StoreEver MSL3040 is selected over the Dell PowerVault ML3 as the LTO-9 tape library on stronger, more directly relevant evidence: HPE's current LTO-9-era product material confirms LTO-9 Ultrium drives, LTO-9 WORM media, and 12 Gb SAS connectivity alongside the slot/drive specification, whereas the retrievable Dell ML3 specification sheet is an older LTO-6/7/8 revision not confirmed to sit in the same document as Dell's separate LTO-9 support note. The selected architecture is a 3U base module, 40-slot class, with 1 × LTO-9 drive initially. The project's tape architecture has no requirement for concurrent tape jobs or drive-level library HA, and a second

drive without a supporting recovery-window calculation would be speculative cost. The 40-slot base module provides sufficient automation capacity for the calculated 32-cartridge minimum GFS media estate without an expansion module, not because 40 “comfortably exceeds” 32: the 32-cartridge figure is calculated retained-media inventory including cartridges expected to be offline/offsite, not a simultaneous in-library occupancy requirement. Exact SKU, PSU configuration, and rated wattage remain a procurement validation item.

Branch Core uses a redundant pair of Catalyst 9200L-family switches (§4.1.4), removing a single core-switch dependency for all 250 Branch employees. This is a genuine TR-005 consideration distinct from the accepted single-unit Backup & Recovery server exception, since that server’s quantity is a fixed constraint while Branch core quantity is not. Confirmed physical data: 1U per unit, dual hot-swappable PSU supported. Manufacturer power consumption varies by the selected Catalyst 9200L SKU, port count, and PoE loading, so the final value is held for UPS sizing until the exact production SKU is fixed. Confirmed physical occupancy is 1U per unit, giving 2U for the redundant Branch Core pair.

Branch Security uses $2 \times$ Meraki MX85 in HA (§4.1.8). A genuine architectural asymmetry exists relative to HQ: the MX85 has a single, non-redundant internal power supply, unlike the HQ MX250 (dual-PSU, HA-capable). Branch resilience is therefore provided at the appliance/system level: two MX85 appliances in HA, each supplied from a different protected power path, so loss of one appliance, its PSU, or its associated power path does not remove the entire Branch security/WAN service by design. Actual failover behaviour remains to be validated against the final Meraki HA configuration. Confirmed physical data: 1U per unit, single non-redundant internal PSU, rated consumption $\approx 12\text{W}$ idle / 55W max: $2 \times 1\text{U} = 2\text{U}$, and a defensible Branch security planning load of $2 \times 55 = 110\text{W}$.

4.4.2 A/B Electrical Power-Path Architecture

Independent A/B UPS-and-distribution paths are selected at both HQ and Branch (a System-plus-System topology, not a single UPS with internally redundant modules). This follows from extending TR-005’s single-device/single-link-dependency principle into electrical infrastructure, which gives clearer failure-domain separation than one UPS chassis with internal redundancy. Each UPS path is independently sized to support the full critical load following failure of the alternate path. Under normal operation, load is distributed across the A and B paths with utilisation monitored to preserve failover capacity; a fixed 50/50 split is not assumed.

Established design parameters:

ID	Decision
PWR-001	Independent A/B online double-conversion UPS architecture at HQ and Branch
PWR-002	Each UPS path independently supports 100% of the defined critical load following failure of the alternate path
PWR-003	Dual-PSU critical equipment connects PSU-A to PDU-A and PSU-B to PDU-B (never both PSUs of one device to the same path)
PWR-004	Single-PSU appliances forming redundant HA pairs are split across A/B paths (e.g., MX85-A $\rightarrow$ PDU A, MX85-B $\rightarrow$ PDU B)

ID	Decision
PWR-005	A fixed normal-operation 50/50 load split is not assumed; path utilisation is monitored to preserve failover capacity
PWR-006	SAN controllers/enclosures receive independent A/B electrical feeds, consistent with their redundant-controller/redundant-storage-path design: completing the layered protection argument: disk protection + controller redundancy + storage-network path redundancy + electrical path redundancy
PWR-007	A single-PSU, non-redundant device with no HA peer is recorded as a residual power dependency, not silently treated as A/B-redundant, unless a validated transfer/dual-input mechanism is introduced
PWR-008	HQ and Branch use the same A/B redundancy principle but are sized independently: $UPS(HQ) \neq UPS(Branch)$ in general

The tape library is not classified as Tier-1 critical load requiring uninterrupted operation through every power-path failure: temporary tape-library unavailability does not take production services offline, since the local disk repository remains available and scheduled tape operations resume after restoration. If the final MSL3040 configuration cannot be independently dual-fed, this is recorded as a lower-tier backup-device power dependency rather than distorting the site-wide A/B architecture around it; its load must still be included in UPS sizing once known. The previously identified Eaton 9PX G2 UPS family (§4.1.11) remains provisional, pending the final watt/VA and runtime calculation: the project sizes first and selects the model second.

4.4.3 UPS Load Evidence Status

Final UPS watt and VA requirements cannot yet be calculated reliably, because manufacturer-rated consumption figures for several selected infrastructure components are configuration-dependent and available only through interactive vendor sizing tools (Dell's Power Calculator is robots.txt-blocked; HPE Power Advisor is a JavaScript-driven configurator) or final procurement configurations. PSU nameplate ratings are not substituted for equipment consumption, since doing so would materially overstate or misrepresent actual load. The current verified load subtotal therefore includes only equipment for which manufacturer-rated maximum consumption has been established.

Device	Evidence class	Treatment
HQ Meraki MX250 pair	A	Use $2 \times 190 = 380W$
Storage-fabric Catalyst 9300X-24Y pair	A	Use $2 \times 243.7 = 487.4W$
Branch Meraki MX85 pair	A	Use $2 \times 55 = 110W$
Catalyst 9200L (Branch Core)	A, SKU-dependent	Value held until exact production SKU fixed

Device	Evidence class	Treatment
Catalyst 9500 / 9300 (HQ Core/Distribution)	C	PSU capacity not accepted as equipment load
Hyper-V hosts, ERP servers, Backup & Recovery server	Evidence Required	Manufacturer configurator at procurement, using the established §4.2 configurations
ME5024 storage array	Evidence Required	Configuration-specific manufacturer value required
MSL3040 + LTO-9 drive	Evidence Required	HPE MSL Power Calculator / procurement confirmation

Partial confirmed-load subtotals (explicitly not total critical load, only the presently evidenced subset): P(HQ, confirmed) = 867.4W, comprising the MX250 HA pair (380W) and the dedicated Catalyst 9300X-24Y storage-fabric pair (487.4W). P(Branch, confirmed) = 110W (MX85 pair only). The Catalyst 9200L is explicitly not summed into the Branch subtotal until its production SKU is selected.

This limitation affects only quantitative UPS capacity selection; the established A/B power architecture remains unchanged, and each UPS path must still be capable of supporting 100% of the final validated critical load. In place of a specific kVA/model selection, the established parametric requirement is: $P(\text{UPS-A}) \geq P(\text{critical, validated})$ and $P(\text{UPS-B}) \geq P(\text{critical, validated})$, independently, per site, with HQ and Branch sized separately per PWR-008. No Eaton/APC model or specific kVA rating is selected at this stage, since doing so from a partial total would convert an evidence gap into an invented engineering result.

4.4.4 Rack Architecture: HQ

HQ rack item	U
Currently validated active equipment (including the 2×1U Catalyst 9300X-24Y storage-fabric switches)	21U
Planned rack-support space (patch panels, horizontal cable management)	4U
Formal 25% growth allowance	7U
Calculated HQ rack requirement	32U

A single 42U floor-standing enterprise cabinet is selected as the HQ baseline, leaving 10U physically unoccupied against the 32U design requirement. The current known HQ equipment subtotal and calculated rack requirement exclude production HQ Distribution and Access switching, because final switch quantities and physical placement depend on endpoint distribution, PoE demand, telecommunications-room locations, and cabling topology, information C-003 confirms is not available. HQ Distribution and Access switches are assumed to be installed in appropriately located telecommunications-room/IDF cabinets serving their respective access areas, separate from the central 42U server/data-centre cabinet, not in the central cabinet itself. The 42U cabinet is therefore a baseline for the currently validated central equipment estate, not evidence that all future HQ access switching will be installed in it. Researched major-device weights (Dell PowerEdge R760 ≈36kg fully populated, PowerVault ME5024 ≈30kg populated, HPE StoreEver MSL3040 ≈35kg with

media/drives) sum comfortably below a representative 42U cabinet’s static-load benchmark, and no U-space, weight, cable-density, heat-density, or physical-room evidence supports splitting into separate network and server/storage racks at the currently known scale. Cabinet depth is not established as a specific dimension: a standard enterprise server cabinet with sufficient usable mounting depth for the deepest selected equipment, rear cabling, vertical PDUs, and service clearance is specified, with a final cabinet depth/SKU deferred until the MSL3040’s depth and manufacturer clearance requirements are confirmed.

4.4.5 Rack Architecture: Branch

Branch rack item	U
2 × Catalyst 9200L core	2U
2 × Meraki MX85 HA pair	2U
Local patch panel	1U
Horizontal cable manager	1U
Known planned occupancy	6U
Formal 25% growth allowance	2U
Known Branch calculated requirement	8U

A provisional 24U floor-standing cabinet is adopted as the Branch baseline, explicitly larger than the 8U calculation, to avoid unnecessary redesign risk given C-003’s unresolved access-switch count and telecommunications-room arrangement for a 250-person site (a 12U cabinet would leave only 4U before the access layer is even known). This is a provisional design baseline, not an exact procurement specification, and is not presented as evidence the Branch needs 24U from the current equipment count alone. Final cabinet size, quantity, and location remain subject to the site survey required by C-003.

4.4.6 Rack Layout Principles

Heavy equipment (ERP servers, the Backup & Recovery server, storage enclosures, and later rack-mounted power/battery equipment) is installed predominantly in the lower portion of the rack. Patch panels and horizontal cable managers are distributed near the equipment they serve, rather than collected remotely at one end. A/B PDUs are rear vertical 0U units where the selected rack supports them, preserving front equipment U-space. Unused rack openings receive blanking panels to reduce hot-air recirculation, though no specific cooling-efficiency percentage is claimed for this practice. The tape library is positioned at a serviceable height for cartridge loading and operator access, not simply high in the rack because it is lighter than the servers. The 25% growth allowance applied throughout §4.4 is explicitly a project engineering allowance, not attributed to ASHRAE, TIA, or any external standard.

4.4.7 Environmental Design

Airflow follows a cold-front, hot-rear orientation and is compatible with the documented airflow direction of the principal shortlisted equipment (the Catalyst 9500’s standard fan modules are front-to-back only, by Cisco’s own design; the Catalyst 9300 supports front intake with rear exhaust across all SKUs; Dell/HPE rack servers are standard front-to-back). Where multiple rack rows are eventually required, conventional cold-aisle/hot-aisle principles apply; exact aisle dimensions and containment arrangements are not specified, since physical room dimensions were not supplied under C-003.

The normal equipment-inlet temperature target is 18°C–27°C, per ASHRAE’s current recommended envelope for Classes A1–A4, distinguished from the wider allowable-range extremes. Environmental monitoring measures rack-inlet temperature and moisture conditions close to IT equipment intake locations, rather than room-level HVAC readings alone, since the relevant equipment thermal environment is the air actually entering the equipment, not room-average or return-air temperature. Moisture conditions are maintained within the applicable ASHRAE recommended envelope, expressed via both relative humidity ($\leq 70\%$, or $\leq 50\%$ where corrosive pollutants are present) and dew point (–9°C to 15°C), with the more restrictive of the two governing: no simple “40–60% RH” figure is adopted as a substitute for this nuance.

Cooling capacity and redundancy are not established as a specific CRAC quantity or N+1 arrangement, blocked by the same unresolved manufacturer configuration-specific power/heat evidence identified in §4.4.3: the 1W IT electrical load $\approx$ 1W heat load conversion principle applies, but the validated IT watts do not yet exist to apply it to. Cooling capacity and redundancy will be selected against the validated IT heat load and the project’s availability requirement once that evidence exists, avoiding an intentional single cooling dependency for critical equipment.

Fire, leak, and physical protection are stated as production engineering recommendations, not Ghana statutory compliance claims: early smoke detection suitable for an IT/electrical equipment environment (aspirating-detection class, consistent with conventional industry practice under the US NFPA 75/NFPA 2001 framework), clean-agent fire suppression where approved by the competent local fire/building authority, appropriate building fire protection as required by applicable local code, water/leak detection around cooling or other potential water sources, restricted physical access to the server/network room, door/access logging where practical, and environmental alerting integrated with monitoring operations. NFPA 75/NFPA 2001 are not claimed as legally mandatory in Ghana: no Ghana-specific building/fire-code primary source was located, and the US-centric source used does not address international applicability.

4.4.8 Remaining Dependencies

§4.4 is architecturally complete for TDR purposes. Final engineering quantities for UPS capacity/runtime and cooling capacity/redundancy remain procurement-stage evidence dependencies because the complete configuration-specific IT power and heat load has not yet been validated. Both dependencies trace to the same unresolved manufacturer configuration-specific power/heat evidence gap, not to any unfinished architectural decision: UPS methodology is closed, sizing is open; the cooling architecture requirement is established, quantity is open.

Section 5: Security Architecture

This section specifies GlobalBridge’s security architecture across five domains: perimeter security, internal network security, wireless security, server and endpoint security, and governance and compliance.

5.1 Perimeter Security

A four-zone model (Internet/DMZ/Internal/Management) is used, with deny-by-default enforced between zones. The DMZ houses only the externally published web-facing tier and any future application services that genuinely require Internet exposure; AD/DNS, DHCP, ERP, backup, SIEM, and internal file services are explicitly excluded from DMZ placement. The HA NGFW pair at each site (Meraki MX250 at HQ, Meraki MX85 at the Branch, per §4.1.7/§4.1.8) provides stateful filtering, application-aware control, and logging. IPS operates inline rather than as passive IDS-only, with vendor-managed signature updates and monitored update health. Anti-DDoS protection is layered: ISP/upstream scrubbing serves as the primary defence against volumetric attacks, since an attack saturating the circuit cannot be solved by the on-premises firewall alone; the NGFW provides

complementary local filtering; and cloud scrubbing remains an optional addition subject to the existing data-residency assessment.

VPN is provided in two forms: site-to-site IPsec per the established Section 3.6 WAN architecture (the Packet Tracer GRE tunnel remains simulation-only, not production encryption), and remote-access SSL VPN with mandatory MFA for employees and administrators, named-identity plus RADIUS/central-identity integration, and device certificates for managed corporate devices.

Packet Tracer boundary: INTERNET-INBOUND and GRE connectivity are PT-verified; NGFW inspection, IPS, DDoS mitigation, SSL VPN, MFA, and production IPsec are not claimed as PT-verified.

5.2 Internal Network Security

The VLAN architecture (§3.3) remains the principal segmentation mechanism. Private VLANs (PVLANS) are used selectively within server-farm segments to restrict unnecessary same-subnet communication while preserving explicitly required peer communication for the File & Print cluster, ERP replication, Hyper-V cluster traffic, and the Wazuh master/worker/indexer topology (§4.2). Production wired access uses IEEE 802.1X with EAP-TLS and RADIUS, with EAP-TLS preferred for managed endpoints as a certificate-based method that avoids reliance on reusable passwords. RADIUS integrates with the existing Active Directory identity environment, and no new commercial RADIUS product has been selected. Unauthorised devices are denied by default or placed in a restricted quarantine/remediation VLAN with no normal internal access.

Layer 2 protections, including DHCP Snooping, Dynamic ARP Inspection, and IP Source Guard, operate as complementary controls rather than substitutes for one another. Spanning-tree protections (PortFast with BPDU Guard on edge ports, Root Guard on appropriate downstream interfaces) are assigned by interface role rather than enabled blanket-wide. Administrative security measures include disabling and removing unused ports from production VLANs, replacing Telnet with SSH, restricting management sources, and applying central AAA with named administrative identities.

Packet Tracer boundary: VLAN segmentation, the BLOCK-FINANCE-TO-HR and MGMT-ACCESS ACLs, management-source restriction, SSH-only VTY, PortFast, and BPDU Guard are PT-verified; 802.1X/EAP-TLS, RADIUS, PVLANS, DHCP Snooping, DAI, IP Source Guard, and Root Guard are not.

5.3 Wireless Security

Three logically separated SSID classes are used: Corporate (WPA3-Enterprise with 802.1X/EAP-TLS, identity/policy-based access), Guest (captive portal, Internet-only, isolated from corporate, management, and server networks, with client-to-client isolation where supported), and IoT (device-appropriate authentication with an explicit allow-list limited to required services, carved out because many IoT devices cannot support corporate-grade authentication or endpoint controls). Actual SSID names are not invented.

Rogue AP detection is provided by wireless management monitoring for unauthorised APs, with detected devices classified and investigated through the security incident process; automatic containment is used only where legally and operationally appropriate, not assumed for every detected neighbouring AP. The Aruba AP-635/Aruba Central platform (§4.1.9/§4.1.10) remains the production baseline; cloud-held configuration, telemetry, and log data remain subject to the existing TR-007 review scoped to client-financial-data storage/backup/replication/recovery (§4.1.10). AP quantity and placement remain C-003/site-survey dependent, unchanged from §4.1.

Packet Tracer boundary: none of Aruba Central, WPA3-Enterprise, RADIUS, or rogue-AP detection is claimed as PT-verified.

5.4 Server and Endpoint Security

A CIS Benchmark hardening baseline applies to Windows Server 2025 and Ubuntu Server 24.04 LTS (§4.2's selected platforms), with deviations documented where operational requirements prevent direct adoption; this is treated as a baseline, not as evidence that every recommendation is already implemented.

A project-selected patch-management SLA (not attributed to CIS or Ghanaian law) applies: critical, actively-exploited or emergency vulnerabilities are remediated within 72 hours where a validated patch or mitigation exists; other critical updates within 7 days; high within 14 days; and medium and lower priority within normal monthly maintenance. Emergency changes are still tested proportionately where operationally possible.

MFA is mandatory for privileged/administrator accounts, remote-access VPN, security-management systems, cloud administrative portals, backup administration, SIEM administration, and other high-risk administrative interfaces where supported, with phishing-resistant methods (FIDO2/security keys, certificate-backed) preferred over SMS. Privileged Access Management (PAM) is treated as a distinct capability, not merely as having administrator accounts: it covers privileged credential vaulting, brokered/check-out access, MFA for privileged operations, just-in-time/time-limited privilege where supported, session and audit recording, credential rotation, separation of normal and privileged accounts, and tightly controlled break-glass credentials. The exact PAM product is left to procurement.

Endpoint controls include centrally managed EDR/anti-malware, host firewalling, disk encryption where appropriate, controlled software installation, and central patch/vulnerability management. Microsoft Defender for Endpoint P2 is the selected EDR solution, with the cost and integration justification addressed in Section 8.

5.5 Governance and Compliance

ISO/IEC 27001:2022 is selected as the principal information-security governance framework, with a reasoned rejection of ITIL 4 (primarily IT service management, not security governance) and acknowledgement of COBIT 2019 as a credible enterprise-IT-governance alternative: ISO 27001:2022 maps most directly to establishing and operating an information security management system. This is explicitly not a certification claim.

A control-mapping table maps twelve controls (SEC-01 through SEC-12: perimeter filtering, internal segmentation, Finance/HR isolation, secure administration, network admission, Layer 2 protection, wireless security, remote access, server hardening, privileged access, security monitoring, recovery protection) to their implementation method and evidence artefact. It cites already-established project elements (Wazuh for SEC-11 security monitoring, the 3-2-1-1-0/offline-offsite-Ghana backup architecture for SEC-12 recovery protection) rather than re-deciding them, and marks "production evidence required" wherever Packet Tracer does not cover the control.

A data classification policy defines four levels, Public, Internal, Confidential, Restricted, with handling rules per level. Restricted explicitly ties to TR-007's Ghana-residency requirement for client financial data, and classification follows the information when copied, exported, or backed up, not the device it happens to sit on. An Acceptable Use Policy summary covers authorised use, credential protection, authorised software/devices, classification compliance, no control bypass, no unauthorised scanning, incident reporting, and remote-access/removable-media compliance, with monitoring notice; violations follow the HR/security disciplinary process, with no invented automatic sanction. The Ghana Data Protection Act 2012 (Act 843) is reaffirmed with source, ISO/IEC 27001:2022 is restated, and PCI DSS remains explicitly conditional: it applies only if cardholder-data activity actually falls within scope, not merely because generic financial information is handled.

Detailed SIEM operations, retention, alerting, and monitoring architecture are addressed in Section 6, not introduced here as a Section 5 domain.

Section 6: Monitoring, Logging & Operations

This section specifies GlobalBridge's operational monitoring, logging, and change-management architecture, covering six areas: Network Management System, SNMP configuration, syslog architecture, SIEM integration, NetFlow/sFlow analysis, and change management process.

6.1 Network Management System (NMS)

LibreNMS is selected as the production NMS, chosen through a structured comparison against PRTG and Nagios:

Criterion	LibreNMS	PRTG	Nagios
Multi-vendor monitoring	Yes	Yes	Yes
SNMP	Yes	Yes	Yes
Syslog integration	Yes	Yes	Available, config/component-dependent
Flow monitoring	NetFlow/sFlow/IPFIX via integration	Strong integrated	Available via associated tooling
Alerting	Yes	Yes	Yes
Distributed/scalable	Distributed polling	Supported, architecture/licensing-dependent	Supported, deployment/product-dependent
Config-backup integration	Oxidized/RANCID	Not the main advantage	Not the main advantage
Licensing	Open-source	Commercial	Core open-source + commercial variants
Selected?	YES	No	No

LibreNMS is selected for how well it matches the project's requirements, not simply because it is open source. It supports the multi-vendor estate (Cisco, Meraki, Aruba, Eaton, Dell, HPE) and brings configuration-backup and syslog visibility together without introducing a commercial licence ahead of the Section 8 cost analysis. This is not assumed to mean zero total cost of ownership: commercial support and implementation costs are addressed in Section 8. The capability claims above are confirmed directly against LibreNMS's own feature documentation: SNMP, syslog (integrated/Graylog), NetFlow/sFlow/IPFIX (via NfSen), distributed polling, auto-discovery, alerting, environmental sensors, STP/OSPF/OSPFv3/BGP data collection, VLAN/ARP/FDB table collection, and Oxidized/RANCID device-backup integration are all explicitly vendor-documented. LibreNMS is the operational NMS; Wazuh (§4.2, §6.4) remains the separate SIEM platform, and the two are not conflated. Final server sizing, support arrangements, and implementation/TCO implications for LibreNMS are validated in Section 8.

Monitored metrics draw on the multi-vendor estate's standard telemetry: link/interface status and utilisation, interface errors and discards, device CPU and memory, environmental sensors

(temperature, power) where the device exposes them, and network control-plane and Layer 2 state, including STP and OSPF, along with other supported protocol telemetry where applicable. This directly supports the availability and capacity-management posture already established under TR-004's 99.999% target and TR-005's no-single-point-of-failure principle (§2.2, §3.1): sustained link or device degradation becomes visible operationally, rather than surfacing only through user-reported outages.

Alerting thresholds are anchored to the already-established TR-010 WAN utilisation figures (§3.6, §2.2) rather than set independently: sustained utilisation approaching the 70% threshold already established for HQ (700 Mbps of the 1 Gbps primary circuit) and Branch (350 Mbps of the 500 Mbps primary circuit) triggers a capacity-planning alert, distinct from a hard outage alert. Device- and interface-level alerting (link down, high error rate, environmental threshold breach) uses LibreNMS's native alerting engine; exact numeric thresholds beyond the TR-010-derived WAN figures are left as an operational tuning parameter rather than presented as fixed values with no real basis. Alerts generated here feed the incident-response and escalation process already established under §5.5's SEC-11 (security monitoring) control mapping and the Section 7 DR notification sequence, rather than defining a separate, disconnected escalation path.

Packet Tracer boundary: interface/link state and STP/OSPF data as visible through native Packet Tracer show commands are demonstrable in simulation; LibreNMS itself, its polling, dashboards, and alerting are not Packet-Tracer-verifiable and are not claimed as PT-verified.

6.2 SNMP Configuration

SNMPv3 authPriv is used for production-capable infrastructure, with SHA-family authentication and AES-family privacy applied where mutually supported by the monitored device and the NMS. SNMP access is restricted to authorised management systems (the LibreNMS polling host and any authorised administrative workstations), consistent with the management-source restriction already established in §5.2. SNMPv1/v2c community strings are not the preferred production management mechanism. A specific SHA/AES variant is not established here, pending confirmed mutual support between each monitored device family and the NMS at implementation time.

A representative MIB set is monitored: IF-MIB (interface status, utilisation, errors), SNMPv2-MIB/system (device identity, uptime), ENTITY-MIB (hardware inventory), ENTITY-SENSOR-MIB where supported (environmental sensors), HOST-RESOURCES-MIB where applicable (host-level CPU/memory/storage), and vendor-specific MIBs for the Cisco, Meraki, Aruba, Eaton, Dell, and HPE equipment selected in §4.1-§4.4.

Packet Tracer boundary: SNMPv3 authPriv configuration itself is not claimed as PT-verified.

6.3 Syslog Architecture

A centralised syslog architecture is used: devices forward to central syslog collection, which provides operational/NMS visibility and simultaneously forwards security-relevant events to Wazuh (§6.4), with the full log stream retained in a protected archive per the retention policy below. This deliberately keeps the operational syslog function distinct from Wazuh's own separate SIEM retention (§4.2, §6.4): they are two related but non-identical retention regimes, not one pool of logs described twice.

Retention policy: operational syslog is retained 90 days online/searchable plus 12 months archived. This is explicitly a project-selected retention policy, not a Ghana Data Protection Act or ISO 27001 mandate: no such duration is prescribed by either source, and none is claimed here.

Log integrity protection: restricted write/delete permissions on the syslog store; administrative separation where practical between those managing the syslog platform and those it monitors; authenticated administrative access only; reliable timestamping and time synchronisation across the

estate; backup of retained logs consistent with their data classification; cryptographic hashing/integrity verification for archived logs where the storage platform supports it; and immutable/WORM-capable archival where justified for higher-value audit records. This last point does not assume the existing backup repository (§4.3, §7) is automatically WORM/immutable: that capability was never established and is not claimed here.

Residency: TR-007's Ghana-residency requirement (§2.2, §4.1.10) governs only Restricted-classification client financial data, not all logs by default. Log residency therefore follows the data classification assigned under §5.5's four-level policy, not a blanket Ghana-residency claim applied uniformly to every operational log.

Packet Tracer boundary: syslog forwarding as demonstrable through Packet Tracer's own logging commands is PT-verifiable in a limited sense; the centralised retention architecture, integrity controls, and residency scoping described here are not.

6.4 SIEM Integration

Wazuh, selected and sized in §4.2 as a 6-VM distributed platform (1 master, 2 workers, 3 indexers), is the SIEM. Log sources feeding Wazuh include the CIS-hardened Windows Server 2025 and Ubuntu Server 24.04 LTS endpoints established in §5.4, network and security device logs relevant to the SEC-01-SEC-12 control mapping (§5.5), and the security-relevant subset of the centralised syslog stream (§6.3) forwarded from perimeter, internal, and wireless infrastructure.

Ten initial correlation-rule categories form the starting operational baseline: repeated authentication failures; a successful login following repeated failures; privileged-group or account changes; off-hours privileged login; account lockout; security-control changes; logging/agent interruption; malware or endpoint alerts; suspicious administrative activity; and critical infrastructure events. Exact thresholds (failure counts, time windows) are left as operational tuning parameters rather than presented as fixed figures with no real basis.

Packet Tracer boundary: Wazuh itself, its correlation engine, and its alerting are not Packet-Tracer-verifiable and are not claimed as PT-verified.

6.5 NetFlow / sFlow Analysis

Flow-based bandwidth and anomaly-detection visibility is provided through LibreNMS's documented NetFlow/sFlow/IPFIX integration (via NfSen), applied across the multi-vendor estate established in §4.1. This serves two related but distinct purposes: bandwidth/capacity trend analysis, feeding the same TR-010-derived utilisation thresholds referenced in §6.1, and anomaly detection, that is, identifying traffic patterns inconsistent with expected baselines (for example, an unexpected sustained flow toward an unfamiliar destination, or a volume spike inconsistent with normal business-hours traffic) that may warrant investigation through the incident process referenced in §5.5 and §6.1. Exact flow-export configuration (sampling rate, exporter placement) is an implementation-stage decision, not established here.

Packet Tracer boundary: NetFlow/sFlow export and analysis are not Packet-Tracer-verifiable and are not claimed as PT-verified.

6.6 Change Management Process

A structured, five-stage change-management process governs modifications to GlobalBridge's production infrastructure, aligned with the ISO/IEC 27001:2022 governance framework already selected in §5.5 rather than introducing an unrelated framework for this one item:

1. **Propose and assess.** Changes are submitted with a description of the change, its business or technical justification, affected systems, expected impact, implementation risk and rollback

plan. The assessment determines the appropriate testing, approval level and maintenance requirements before implementation begins.

2. **Test.** Non-emergency changes are validated in a non-production environment before touching production infrastructure: for configuration changes within the scope of this project's Packet Tracer topology, this is the same PT environment used throughout Sections 3-5; for changes outside that scope (server, application, or cloud-service changes), an equivalent staging/lab validation applies. This mirrors the patch-management testing principle already established in §5.4 (emergency changes are still tested proportionately where operationally possible), rather than introducing a separate, disconnected testing standard.
3. **Approve.** Changes are authorised before implementation, with authority proportionate to risk: routine, low-risk changes may be approved by the responsible system owner; changes affecting Tier 1 services (§2.2, §7), security controls (§5.5's SEC-01-SEC-12 mapping), or multiple systems require review by IT management, consistent with the named administrative-identity and accountability principle already established in §5.2 and §5.4.
4. **Implement, with defined rollback.** Approved changes are implemented within a scheduled maintenance window where the change is not emergency-classified, using the rollback plan captured at proposal. Before a network-device configuration change, the current configuration is backed up and its availability confirmed using the Oxidized/RANCID configuration-backup integration identified in §6.1. This provides a known pre-change configuration that can support rollback if the implementation fails.
5. **Verify and record.** Post-implementation verification confirms the change achieved its intended effect without unintended side effects on monitored metrics (§6.1) or security posture (§5.5); the change and its outcome are recorded for audit traceability, consistent with the evidence-artefact discipline already established in §5.5's control-mapping table.

Emergency changes, for example an active-exploitation patch under the 72-hour SLA established in §5.4, follow the same assessment, approval, implementation and verification principles on a compressed timeline. Testing remains proportionate where operationally possible, and approval is obtained from an available authorised approver rather than waiting for a scheduled review cycle.

Packet Tracer boundary: the change-management process itself, as an organisational/procedural control, is not Packet-Tracer-verifiable; only the underlying configuration-backup and test-environment mechanics it references (Oxidized/RANCID, the PT topology as a test environment) have any PT-adjacent evidence, and even those are not claimed as direct verification of the process itself.

Section 7: Backup, Disaster Recovery and Business Continuity

7.1 Backup Strategy

Recovery objectives and tiers. The backup strategy is built directly on the three-tier recovery framework already established under TR-006 (§2.2): Tier 1 critical services (ERP database, AD/DNS) carry a 30-minute RTO and 15-minute RPO; Tier 2 important services (File & Print, DHCP, Web/Intranet) carry a 4-hour RTO and 1-hour RPO; and Tier 3 standard services (SIEM/logs, provisional) carry a 24-hour RTO and 24-hour RPO. RPO achievement and 3-2-1-1-0 protection are treated as separate mechanisms, not as one backup job doing both: the offline/offsite tape layer is not responsible for meeting Tier 1's 15-minute RPO on its own. That RPO is met through an operationally separate mechanism (application-aware transaction-log backups plus replica), consistent with vendor guidance on typical SQL Server RPO practice.

Backup platform and architecture. Veeam Backup & Replication, already selected in §4.3 as the reference implementation for the tape/GFS architecture and referenced again in §4.4's repository sizing, is the production backup platform. Veeam provides inline deduplication and compression during backup-job processing, and can additionally leverage ReFS fast-clone on the established 64

KB-allocation-unit ReFS repository (§4.3) to reduce the physical space consumed by synthetic full backups and retained restore points. The established ≥ 51 TB usable-repository requirement remains a physical-capacity floor: none of Veeam's deduplication, compression, or fast-clone savings are credited toward that figure, since it was derived (§4.3) from the full logical backup baseline plus a 14-day change-rate window sized as if no software-level space reduction occurs at all. Any space savings realised in production are additional headroom beyond the established requirement, not a substitute for it.

The architecture follows the 3-2-1-1-0 rule: three copies of the data (production plus two backup copies), across two different media types (disk repository and LTO-9 tape), one copy offsite (Ghana-resident tape rotation), one copy offline/air-gapped (the tape tier, physically disconnected from the network), and zero errors, with backups verified as described under backup monitoring and verification below. The disk repository (≥ 51 TB usable, §4.3) covers the 14-day operational recovery window, while the LTO-9 tape tier covers the longer-term Grandfather-Father-Son (GFS) generations. This disk/tape split means the full GFS retention policy does not need to fit on local disk at the same time.

Backup schedule per data type. Backup cadence follows the established recovery classification. The ERP database (Tier 1) receives a weekly full backup, a daily differential backup, and a transaction-log backup every 15 minutes to meet its RPO. AD/DNS (Tier 1) receives application-aware/system-state protection appropriate to the deployed Windows platform, distinct from the SQL transaction-log methodology since that approach does not apply to AD. File & Print, DHCP, and Web/Intranet (Tier 2) receive a weekly full backup plus hourly incremental backups. SIEM/logs (Tier 3, provisional) rely on Wazuh's own retention mechanism (§4.2, §6.4) for its searchable dataset, with the conventional backup architecture protecting only Wazuh's configuration, rules, dashboards, and integrations, not the full searchable log dataset.

Retention policy. The GFS retention policy is 14 daily, 4 weekly, and 12 monthly recovery points. This is an explicitly declared project policy, not one sourced from 3-2-1-1-0 or the Ghana Data Protection Act 2012: neither specifies a retention count or period, and the Act's own text (Section 24) requires only that a controller justify and observe its own retention period, not that it use a specific duration. This policy may be revised if a specific contractual, financial, ERP, or regulatory retention requirement is later identified for a particular dataset. The weekly and monthly GFS generations are held as independent full recovery points on LTO-9 tape: at 22.918 TB of conventional protected data per generation (2 cartridges of 18 TB native capacity each, rounded up), the 16 retained generations (4 weekly + 12 monthly) require a minimum calculated GFS retention-media baseline of 32 LTO-9 cartridges. This is the mathematically derived minimum, not a claim that GlobalBridge requires exactly 32 tapes: actual procurement may exceed this due to scratch/working media, failed or retired cartridges, cleaning media, platform-specific tape-job behaviour, rotation overlap, and growth above the modelled baseline.

Encryption of backup data. At rest, LTO-9 cartridges use native hardware encryption at AES-256 in GCM mode, implemented at the tape drive controller. This has been a standard LTO feature since LTO-5, confirmed against the LTO Program's own encryption documentation, and it applies to all tape media used for the weekly/monthly GFS tier and for offsite-transported cartridges. Encryption keys are managed separately from the media they protect: keys are not stored alongside, or transported together with, the cartridges themselves, and the specific key-management platform is left as a procurement-stage decision. In transit, Veeam Backup & Replication encrypts backup network traffic transferred over WAN/public-network links by default, using TLS as the underlying transport-security protocol: TLS 1.3, including mutual TLS (mTLS), by default, with TLS 1.2 (including mTLS) supported as a fallback for components that cannot yet negotiate TLS 1.3, confirmed against Veeam's own Help Center documentation. This mechanism must be enabled for any backup traffic crossing a public or otherwise untrusted network segment, specifically including traffic on the offsite/Ghana-resident storage path. No specific cipher suite or exact TLS-configuration

parameter beyond “TLS 1.3 by default / TLS 1.2 as fallback” is established, since that level of detail belongs to configuration time rather than to the architecture.

Offsite/cloud backup destination. The offsite copy required by 3-2-1-1-0 is satisfied by rotating LTO-9 cartridges to a Ghana-based offsite media-vault facility, consistent with TR-007’s residency requirement for client financial data (§2.2, §4.1.10). The offsite destination is therefore a physical Ghana-resident vault rather than a cloud backup destination, avoiding the residency problem already identified for hyperscaler cloud services in §4.2’s Ghana-residency revalidation. The exact offsite media-vault provider is not yet selected (see “Remaining open dependencies,” below); Equinix AC1/Accra is cited only as evidence that suitable Ghana infrastructure exists, not as a selected service.

Backup monitoring and verification. Backup verification and the test-restore schedule are addressed together with disaster-recovery testing in §7.2 (DR Testing), since the two share one testing philosophy, recording corrective-action items rather than retroactively adjusting targets, even though they verify different things.

7.2 Disaster Recovery Plan

DR site strategy and model selection. Four DR site models were compared against GlobalBridge’s actual requirements:

Model	Description	Fit for this project
Hot standby	Fully duplicated, continuously running production-equivalent environment, near-zero RTO	Rejected: would require duplicating the entire estate at full production capacity, disproportionate to Tier 2/3 recovery objectives and not justified given the ≤\$850,000 budget ceiling (C-001)
Cold standby	Recovery infrastructure exists only as bare capacity, with build-out and restoration only after a disaster is declared	Rejected for Tier 1: recovery from bare infrastructure typically takes hours to days, incompatible with the 30-minute Tier 1 RTO/15-minute RPO
Cloud DR (AWS/Azure/GCP)	Recovery environment hosted in a public-cloud region, often via DRaaS	Rejected as the primary strategy: TR-007’s Ghana-residency requirement is not currently satisfiable by the major hyperscalers’ Africa-region footprint (only South Africa is an eligible African geography for at least one major provider’s advanced-residency programme); a foreign-region cloud DR site would recreate the same residency problem already found for Email/M365

Model	Description	Fit for this project
Warm standby	Pre-positioned, reduced-capacity recovery infrastructure, kept partially synchronised, activated and scaled up during a declared disaster	Selected: fits the 30-minute Tier 1 RTO without carrying hot standby's full duplicate-production cost

Selected: warm standby, in a geographically separate Ghana-based facility. This satisfies TR-007, avoids the cost of a fully duplicated production environment, and is achievable within the 30-minute Tier 1 RTO because Tier 1 pre-positions capacity rather than building from a cold state. The exact Ghana facility/provider is not yet selected.

Recovery Time Objective (RTO) and Recovery Point Objective (RPO) per tier. Tier 1: 30-minute RTO, 15-minute RPO. Tier 2: 4-hour RTO, 1-hour RPO. Tier 3: 24-hour RTO, 24-hour RPO, all as established under TR-006 (§2.2).

Tier-dependent recovery mechanism. Warm standby describes the facility's readiness state, not a uniform replication policy applied to every workload. Tier 1 (ERP/database, AD/DNS) justifies pre-positioned recovery capacity and appropriate replication/protection at the warm site, sufficient to meet the 30-minute RTO/15-minute RPO, because rapid backup restoration alone cannot reliably hit a 15-minute RPO for these services. Tier 2 and Tier 3 (File & Print, DHCP, Web/Intranet, SIEM) rely more heavily on backup restoration onto standby infrastructure at the warm site, activated and built out further only once a disaster is declared, consistent with their 1-hour/24-hour RPOs: these tiers do not require continuous replication. This keeps the DR site strategy aligned with the already-established RTO/RPO model and avoids quietly creating a second fully duplicated production environment.

Recovery priority and failover procedure (high-level runbook). Recovery sequencing follows tier order: Tier 1 services are brought up first at the warm-standby facility from pre-positioned capacity and replication, since these carry the tightest RTO/RPO and the greatest business impact; Tier 2 services follow, restored from the backup repository onto standby infrastructure; and Tier 3 services are restored last, consistent with their 24-hour RTO. At a high level: (1) an incident is confirmed as meeting the criteria for a DR declaration; (2) the IT Manager/DR Coordinator declares DR and activates the notification sequence below; (3) Tier 1 services are failed over to the warm-standby facility using pre-positioned capacity/replication; (4) Tier 2 and Tier 3 services are restored from backup onto standby infrastructure, in tier order; (5) service availability and data integrity are verified against the applicable RTO/RPO targets before recovery is declared complete; and (6) once the primary site is restored, a planned failback returns services from the warm-standby facility to primary production infrastructure in the reverse tier order, with verification at each step, to avoid a second uncontrolled cutover.

DR testing (and backup restore verification). Backup restore testing and disaster-recovery exercises serve different purposes. Representative backup restores should be tested quarterly to verify that protected data and systems can actually be recovered. Disaster-recovery preparedness should be exercised through quarterly tabletop exercises, semiannual partial failover exercises, and an annual full DR exercise. Test results should compare actual recovery performance against the applicable RTO and RPO targets. Where an objective is missed, the result should be recorded as a corrective-action item rather than used to change the target retrospectively. All four cadences (quarterly restore test, quarterly tabletop, semiannual partial failover, annual full DR exercise) are project-selected operational policy, not prescribed by the case study, which asks only that a testing schedule exist and names the partial-failover-test category without prescribing frequencies.

Business continuity. During a declared DR event, Tier 1 services are prioritised for restoration ahead of Tier 2/3 to preserve core business operation, consistent with the recovery-priority sequencing above. Interim/manual procedures for affected staff are communicated as part of the notification sequence below rather than left implicit, and any service operating from the warm-standby facility is treated as a temporary state to be resolved through the planned failback procedure once the primary site is restored, not as a permanent operating posture.

Roles and responsibilities.

Role	Responsibility
IT Manager / DR Coordinator	DR-declaration authority; initial escalation; overall coordination of the recovery effort
Network recovery team	Restoring WAN/LAN connectivity, firewall, and routing services at the warm-standby facility
Systems recovery team	Restoring hypervisor hosts, VM infrastructure, and core services (AD/DNS, File & Print, DHCP)
Database/Application recovery team	Restoring the ERP database and dependent applications, verifying transaction-log recovery against the Tier 1 RPO
Security recovery team	Restoring security controls (NGFW, SIEM, monitoring) at the recovery site and confirming no security posture is silently degraded during failover

Communication plan. A concrete, ordered notification sequence governs communication during a DR event:

1. **Incident discoverer / NOC/SOC:** first detection or report, via monitoring/alerting (§6) or direct report.
2. **IT Manager / DR Coordinator:** initial escalation and DR-declaration authority.
3. **Technical recovery teams:** Network, Systems, Database/Application, Security (per the roles table above), mobilised once DR is declared or imminent.
4. **Executive / business owners:** informed of scope, expected impact, and recovery timeline.
5. **Affected staff:** informed of service impact and any interim/manual procedures.
6. **External providers:** carriers, vendors, and the Ghana-based DR/offsite-recovery provider, engaged as required by the recovery procedure.
7. **Clients / external stakeholders:** notified only where contractually or legally required, and only through an authorised communication channel.

Channels are matched to purpose: the monitoring/ticketing system (§6) for initial detection, tracking, and status; telephone, SMS, or an approved secure messaging channel for urgent escalation, particularly steps 2 to 3, where immediate acknowledgement is required; email for formal status updates, documentation, and post-incident reporting where near-real-time urgency is not the primary need; and a predefined alternate channel (a maintained personal/mobile contact list independent of corporate email/ticketing), used specifically if corporate communication systems are themselves unavailable or compromised as part of the incident, so the notification chain does not depend entirely on the systems that may have caused it.

Remaining open dependencies. Two legitimate, deliberately unresolved procurement-level dependencies remain: (1) the exact Ghana-based offsite media-vault provider for tape storage; (2) the exact Ghana-based DR facility/provider for the warm-standby site. Neither requires reopening any technical architecture decision: both are named, scoped procurement dependencies, consistent with how the project has treated equivalent items throughout (SAN/UPS/cooling SKUs, ERP OS, Windows Server edition).

Packet Tracer boundary: no content in Section 7 is Packet-Tracer-verifiable. The backup architecture, encryption mechanisms, DR site strategy, testing regime, and communication plan are all architecture/policy-level content outside Packet Tracer's simulation scope.

Section 8: Budget Summary and Cost Justification

This section presents GlobalBridge's budget summary and cost justification: a professional CapEx/OpEx budget table using real or realistic vendor list prices, realistic reseller discounts (20–40% off list), and confirmation that the Grand Total does not exceed the USD \$850,000 ceiling, with trade-off decisions documented if the ceiling is exceeded, or the reasoning for retaining any significant headroom if it is not.

8.1 Budget Ceiling and Scope Ambiguity

The case study's scenario narrative describes the \$850,000 figure as covering infrastructure only, excluding recurring ISP and cloud service costs. Its own Section 8 instruction, however, requires confirming that **Grand Total = CapEx + Year-1 OpEx ≤ \$850,000**, a stricter reading than the scenario wording alone would suggest. This project follows that explicit, stricter Section 8 instruction as the controlling requirement (CapEx plus Year-1 OpEx, including recurring items such as WAN circuits, licensing, and support, tested against the \$850,000 ceiling), while acknowledging the scenario's narrower framing rather than silently resolving the difference one way.

8.2 Quantity-Status Framework

Every line item in the budget carries one of three explicit statuses, so a reader can distinguish a figure driven by completed design work from one that is a disclosed planning estimate: **Frozen**, where quantity and model are supported by the completed Section 4 architecture; **Planning (C-003)**, where quantity is calculated for budgeting purposes from disclosed capacity-based evidence (VLSM host projections, vendor RF design guidance, or a stated redundancy floor) and is explicitly subject to site-survey or RFQ validation before procurement, never an invented per-employee or per-square-foot ratio; and **Planning allowance**, where the figure (UPS, DR facility, tape-vault) is a budgetary sizing placeholder because final technical sizing or provider selection remains a genuine open dependency carried from Sections 4 or 7.

Planning quantities and their derivation are as follows. HQ Access switches (20× Catalyst C9200-48P) are derived from the established VLSM three-year HQ host projection (800 hosts, VOIP's 300 hosts excluded on the disclosed assumption that IP phones daisy-chain through the PC's access port rather than consuming a separate port), at a 1.15× disclosed spare-capacity policy, divided by 46 usable ports per switch (48 less 2 reserved uplink ports), yielding 20 switches. HQ Distribution switches (2× Catalyst C9300-24T-A) are held at the minimum redundant-HA-pair floor required by TR-005, not a port-count formula, and are explicitly disclosed as likely to increase once real MDF/IDF/floor topology is known. Branch Access switches (9× Catalyst C9200-48P) use the same VLSM-based method against the Branch's established 350-host three-year total. Wireless APs (32× Aruba AP-635) are derived from Aruba's own RF design guidance (30–40 concurrent associated clients per AP) applied against the case study's device-growth ratio (+40.8%) over current headcount, a capacity-based estimate, with final placement still subject to RF site survey. UPS quantities (Eaton 9PX 6kVA×2 HQ, 3kVA×2 Branch) are a budgetary sizing allowance, not derived

from the still-partial confirmed power load established in §4.4, and are explicitly not final engineering sizing.

8.3 Value-Engineering Decision: HQ Access Layer (Catalyst 9300 → Catalyst 9200)

The original §4.1 family-level selection placed HQ Access on the Catalyst 9300 family primarily for single-vendor consistency with the Core, without an HQ-specific comparison against the Catalyst 9200 already selected for Branch Access. It is reviewed here as a genuine equipment-substitution decision rather than a budget cut.

Criterion	Catalyst 9300-48P	Catalyst 9200-48P	Selected?
Wired user access	1 GbE, PoE+	1 GbE, PoE+	Both meet TR-010
Uplink capability	Modular 10/25G uplinks	Modular 10G uplinks	Both meet TR-010's ≥10 Gbps aggregation requirement
Stacking	StackWise-480 (480G/stack)	StackWise-160 (160G/stack)	9300 higher, but no established requirement invokes this figure
SD-Access fabric-edge readiness	Cisco's recommended platform	Model-dependent (§4.1.3)	9300 advantage, but SD-Access is not part of this project's architecture anywhere in Sections 3–5
List price (48-port PoE+, estimate)	≈\$14,500	≈\$6,000	C9200-48P selected

The Catalyst 9200-48P was selected on value-engineering grounds: it satisfies every established HQ Access requirement (1 GbE, PoE+, ≥10 Gbps uplink per TR-010) without paying for stacking bandwidth and SD-Access fabric-edge readiness that no established decision in this project uses, and it gives HQ and Branch Access a single common platform family. The 20-switch **quantity** (Planning, C-003) is unchanged; only the **model family** changed. This substitution reduces discounted hardware CapEx by \$119,000 (a \$170,000 list-price difference across 20 switches, at the 30% reseller discount applied uniformly to hardware), which then cascades through the 10% professional-services, 10% contingency, and 8% access-tier support percentages. The full cascade accounts for the \$153,510.00 total Grand Total reduction this single substitution produces (§8.6).

8.4 Software and Licensing Decisions

Email service: Exchange Online Plan 1 (not Microsoft 365 E3) is the Section 8 budget baseline: E3 lists at \$39.00/user/month (\$351,000/yr for 750 users) against Exchange Online Plan 1's \$4.00/user/month (\$36,000/yr). An Email Server/service role is required, not an enterprise productivity suite: E3's additional Word/Excel/Teams/SharePoint/OneDrive capabilities were never established as a project requirement, and the \$315,000/yr difference would be indefensible as necessary project expenditure. This is a budget-modelling baseline only: cloud-hosted Exchange remains conditional on confirming the final service arrangement satisfies TR-007 for any client financial data stored, processed, backed up, or recovered through the email service, consistent with the Ghana-residency conditional status already established in §4.2.

Windows Server 2025 edition: Standard, selected over Datacenter. Standard requires 3 OSE-license stacks per host $\times 2 \times 16$ -core packs/host $\times 2$ hosts = 12 packs $\times \$729.99 = \$8,759.88$; Datacenter requires 1 stack/host $\times 2$ packs $\times 2$ hosts = 4 packs $\times \$3,699.99 = \$14,799.96$. Standard is cheaper at this VM density (5 of 13 VMs are Windows): Datacenter’s unlimited-VM economics only pay off at materially higher Windows-VM-per-host counts than this design has.

Veeam Backup & Replication: Advanced edition (\$350/workload/yr) is selected over Standard (\$250/workload/yr), because Section 7’s warm-standby DR model requires replication capability Standard does not include. 15 protected workloads (13 VMs plus the 2 physical ERP servers; the Backup & Recovery server itself is infrastructure, not a protected workload) $\times \$350 = \$5,250.00/\text{yr}$.

Endpoint security (EDR): Microsoft Defender for Endpoint Plan 2, selected via a named three-way comparison: Defender for Endpoint P2 \$62.40/endpoint/yr, SentinelOne Singularity Core \$69.99/endpoint/yr, CrowdStrike Falcon Pro \$99.99/endpoint/yr. It is the cheapest of the three, and integrates natively with the Windows Server 2025 estate while also covering the Ubuntu/Linux VMs. 1,150 endpoints (the same VLISM-derived HQ 800 + Branch 350 device population used for switch sizing) $\times \$62.40 = \$71,760.00/\text{yr}$. This is presented as an example of disciplined costing under budget pressure: replacing an earlier unsupported \$60/endpoint placeholder with this sourced \$62.40 figure *increased* modelled cost by \$2,760/yr, since the number was not adjusted downward to help the budget.

Tiered hardware support. A flat 15%-of-discounted-hardware support allowance was replaced with a criticality-tiered model: 18% premium support on Tier-1/critical infrastructure (core, distribution, storage-fabric switches, NGFW, branch core/firewall, UPS, all servers, SAN, tape library) and 8% standard support/warranty on access-layer infrastructure (HQ/Branch access switches, wireless APs). The tiered-support policy initially reduced Year-1 hardware support from \$70,171.61 to \$56,813.39 against the pre-value-engineering hardware baseline, producing the \$13,358.22 Stage 1-to-Stage 2 reduction shown in §8.6. After the later Catalyst 9200 access-layer substitution reduced the underlying hardware value, the same tiered-support policy produced the final Year-1 support figure of \$47,293.39 reported in §8.7.

8.5 WAN Costing Correction

The initial WAN estimate priced all four circuits (HQ primary/secondary, Branch primary/secondary) at equivalent premium dedicated-fibre grade, overstating the two secondary links. TR-010’s actual wording requires secondary connectivity to provide only “at least 50% of the primary link capacity to maintain priority business services if the primary connection fails,” a capacity floor for Tier-1 failover traffic, not parity with the primary circuit’s commercial grade, consistent with the established §3.6 architecture’s “diverse secondary” path. Repriced: HQ secondary \$2,250/mo $\rightarrow$ \$1,000/mo; Branch secondary \$1,350/mo $\rightarrow$ \$600/mo (both still meet or exceed the 50% floor); both primaries unchanged as premium dedicated circuits. This saves \$24,000/yr with no requirement weakened. All four circuit prices remain fully-estimated figures: no Ghanaian carrier publishes enterprise circuit-level pricing, so RFQ engagement with candidate providers (per §3.6.7) is required before procurement.

8.6 Trade-Off Progression

Stage	Grand Total	vs. \$850,000
Initial full-featured design	\$924,388.25	+\$74,388.25 (8.8% over)
After tiered hardware support (18%/8%)	\$911,030.03	+\$61,030.03 (7.2% over)

Stage	Grand Total	vs. \$850,000
After sourced EDR figure (Defender P2, replacing placeholder)	\$913,790.03	+\$63,790.03 (7.5% over)
After WAN secondary-circuit repricing	\$889,790.03	+\$39,790.03 (4.7% over)
After HQ Access value engineering (Catalyst 9300 → 9200)	\$736,280.03	–\$113,719.97 (13.4% under)

Each stage is a documented, evidence-based trade-off, not an arithmetic adjustment: tiered support changes support *level* by criticality, not coverage (§8.4). The EDR step is presented even though it *increased* cost, because it replaced a guess with a sourced figure. The WAN correction is a genuine re-reading of TR-010’s actual text, not a discount. The Catalyst 9200 substitution removes only capabilities (stacking bandwidth, SD-Access readiness) that no established requirement invokes, while explicitly preserving the C-003 planning quantity (20 HQ / 9 Branch switches) untouched.

8.7 Final Budget Position

Budget measure	Amount
Budget ceiling	\$850,000.00
CapEx	\$431,696.87
Year-1 OpEx	\$304,583.16
Grand Total	\$736,280.03
Unallocated budget headroom	\$113,719.97 (13.4%)
Budget utilised	86.6%
Budget status	WITHIN BUDGET

CapEx breakdown: discounted hardware \$348,810.74 (list \$498,301.05, less a disclosed flat 30% reseller discount applied uniformly to physical hardware only, within the specified 20–40% range; software/subscription licensing is priced at list, since equivalent reseller discounting is not standard practice for perpetual/subscription software) + Windows Server 2025 Standard licensing \$8,759.88 (perpetual, undiscounted) + professional services/installation \$34,881.07 (10% of discounted hardware, a disclosed project policy consistent with this project’s other percentage policies: 25% growth reserve, 80% utilisation ceiling, 15% spare port capacity) + contingency \$39,245.17 (10% of CapEx before contingency, a disclosed project policy).

Year-1 OpEx breakdown: tiered hardware support \$47,293.39; Meraki MX250 Advanced Security licence \$14,718.67; Meraki MX85 Advanced Security licence \$4,905.10; Aruba Central subscription \$2,856.00; Veeam Advanced \$5,250.00; Exchange Online Plan 1 \$36,000.00; Microsoft Defender for Endpoint P2 \$71,760.00; WAN circuits (HQ primary \$45,000 + HQ secondary \$12,000 + Branch primary \$31,800 + Branch secondary \$7,200) = \$96,000.00; Ghana DR warm-standby facility allowance \$22,200.00; Ghana offsite tape-vault allowance \$3,600.00.

8.8 DR/Warm-Standby Costing

Only a facility/service allowance (\$1,850/month) and an offsite tape-vault allowance (\$300/month) are included for the Section 7 DR arrangement. No separate duplicate server or storage CapEx is included in this budget. The exact Ghana-based DR facility/provider and the commercial arrangement required to provide sufficient Tier-1 compute, storage and replication capacity remain procurement dependencies. The allowance must therefore be validated through RFQ before procurement, and any final service must satisfy the recovery requirements established in Section 7 without causing the approved budget ceiling to be exceeded.

8.9 Headroom Justification

The design already meets the established capacity, security, resilience, recovery, and three-year growth requirements set out in Sections 2–7; the \$113,719.97 (13.4%) headroom is deliberately not allocated merely because it remains available. Additional expenditure was considered and rejected on evidence: reverting HQ Access to Catalyst 9300 would add stacking capacity and SD-Access readiness that no established requirement uses; increasing WAN capacity, duplicating lower-tier infrastructure, or purchasing uniformly premium hardware support across all equipment would increase cost without a demonstrated requirement. Headroom is retained instead as disclosed tolerance against remaining commercial/technical uncertainty: C-003 still governs the final access-switch, wireless, and UPS quantities pending site survey; Ghana enterprise WAN pricing, the DR facility, and the offsite media-vault service all remain RFQ-dependent; and several server configuration prices are estimates rather than binding quotations. The \$113,719.97 is explicitly labelled unallocated budget headroom, not a second contingency line: the formal \$39,245.17 contingency already sits inside CapEx, and conflating the two would double-count contingency. Any use of the headroom is subject to the project’s change-control process, not this budget summary.

8.10 Discount and Estimate Transparency

Most hardware line items carry a cited source and a real current list price (Cisco/reseller pricing, HPE, Dell component pricing). A smaller set are explicitly labelled ESTIMATE rather than a scraped exact quotation, disclosed at the point of use rather than blended in invisibly: the Dell PowerEdge configured-server prices (Dell’s interactive configurator could not be scripted for the exact single-socket/192 GB builds, so bottom-up component estimates were used instead), the Catalyst C9200-48P and C9300-24T uplink-module prices, the Eaton 9PX 3 kVA unit, the PowerVault ME5024 base-chassis anchor price, and all four Ghana WAN circuits plus the Ghana DR facility/tape-vault allowances (no African enterprise carrier or colocation provider publishes circuit/rack-level pricing).

Packet Tracer boundary: no content in Section 8 is Packet-Tracer-verifiable. Budget, pricing, and cost justification are entirely outside Packet Tracer’s simulation scope.

Appendix A: AI Tool Disclosure

Academic Integrity & Disclosure Rationale

- **Standard Applied:** Fully discloses all AI assistance, including reasoning, research, auditing, and calculations that go beyond basic language support, while confirming that all core architecture decisions, conclusions, and configurations were originated, reviewed, and finalized by the student.

Tool 1: Claude / Claude.ai

Primary Role: Collaborative reasoning partner, consistency auditor, and document maintainer across the Technical Design Report (TDR) and Packet Tracer verification.

Key Use Cases:

- **Reasoning & Review:** Evaluated trade-offs and surfaced conflicts, for example resolving the tension between inbound ICMP blocking and ISP ping verification, and tracing cascading CapEx budget impacts.
- **Consistency Auditing:** Conducted cross-section, rubric, and TDR-versus-Packet-Tracer audits (16 check areas) to identify discrepancies.
- **Evidence Assessment:** Checked Configuration Verification (CV) screenshots, TTL path lengths, and ACL counter deltas against the VLSM plan.
- **Calculations & Research:** Verified the Section 8 budget model progression (from \$924k down to \$736k) and retrieved primary vendor documentation, including Veeam TLS, LTO-9 encryption, Windows Server 2025 SKUs, and EDR pricing.
- **Project Maintenance:** Formatted decision logs and section checklists, and generated .drawio diagram layouts based on the finalized design choices.
- **Organizational & Formatting Support:** Claude supported the arrangement of student-provided data into structured tables, the application of numbering schemes, and the enhancement of document presentation and formatting. All underlying technical specifications, architectural decisions, and configuration instructions were originated, reviewed, and finalized by the student. Claude's involvement in these tasks was limited to structural and stylistic assistance rather than the autonomous creation of technical design content.

Critical Evaluation: No AI output was accepted at face value. All claims were cross-referenced against primary vendor sources, project logs, or CLI show commands.

Tool 2: ChatGPT (OpenAI)

Primary Role: Secondary support tool across later project stages (Sections 3.6 through 8) for troubleshooting, cross-checking, and market research.

Key Use Cases:

- **Research & Comparison:** Researched WAN alternatives (SD-WAN versus MPLS versus DIA) and Ghana ISP options (Comsys, MTN Ghana).
- **Packet Tracer Troubleshooting:** Helped interpret CLI outputs and debug network protocols (GRE, ACLs, DHCP, STP, EtherChannel).
- **Evidence Interpretation:** Assisted with reading CLI outputs and structuring the Configuration Verification Appendix.

Critical Evaluation: Claims surfaced by ChatGPT were independently verified against primary sources through Claude and direct student review. Specific errors were caught and corrected, including an invalid vCPU-to-physical-core calculation, tracking parameters left in URLs, and unsupported claims that were discarded.

Tool 3: Additional Software Tools: diagrams.net (draw.io) Primary Application: Utilized for assembling and styling the visual network topology diagrams. All core technical elements, including structural design, interconnection details, and interface specifications, were established directly through student architectural determinations. The application operated purely as a graphical rendering tool and had no impact on the underlying system engineering. **diagrams.net (draw.io)** Primary use: Construction and visual formatting of the network architecture diagrams. The technical content, including topology, device relationships, and link specifications, originated from student design decisions. The tool served strictly as a presentation and layout aid and did not influence the underlying network architecture.

Key Takeaway

Both AI tools served strictly as analytical and auditing aids. Primary evidence, including Packet Tracer CLI captures and vendor documentation, together with final design ownership, remained entirely with the student.